

**KURYENTA: SOLAR-POWERED RENTAL STATION WITH
DETACHABLE POWER SOURCES**

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Electricity is essential to everyday living, powering household activities from lights and appliances to connectivity. It is the most used energy source and a notable expense among Filipino households, with a 94.8% usage rate (Philippine Statistics Authority, 2023). Frequent power interruptions disrupt comfort, work, and livelihoods, forcing households to rely on diesel generators as temporary backup. However, these generators are costly and contribute to noise and air pollution, highlighting the need for alternative, sustainable power sources.

Research has proven that solar-powered systems offer a reliable and eco-friendly power source. Catalan et al. (2023) developed a solar-powered coin-operated mobile charging station designed that provides off-grid charging for mobile devices through photovoltaic panels and integrated battery storage. Their study emphasized the system's effectiveness in extending energy access to remote and island communities, particularly in locations where electricity is limited. Moreover, Catalan et al. (2023) also emphasized the importance of

incorporating additional security features to ensure the consistent and reliable operation of such systems over time. Similarly, photovoltaic (PV) solar energy is becoming a key player in the global energy transition, providing renewable and scalable energy solutions (Sampaio & González, 2017; Ramly, Jamal, Abd Ghafar, & Babu, 2019).

Technology also plays a vital role in improving such systems. According to Villamil et al. (2020), the Internet of Things (IoT) is an emerging technology present in many devices and processes, improving quality of life. Integrating IoT with solar energy systems enables smart monitoring, management, and accessibility. Rental-based systems can further improve affordability and accessibility. Thakur (2021) illustrated that a car rental system efficiently manages transactions, user data, and resources through database-driven modules and web-based interfaces. Applying similar principles to energy rentals allows automation of verification, monitoring, and payment processes through an IoT-based application. In support of this, Gavera et al. (2024) highlighted the relevance of the ESP32 as a powerful microcontroller for developing efficient and stable sensor networks, enhancing sustainability and resource efficiency. Combining these innovations can maximize utilization, enhance system functionality, and promote sustainable energy access for communities affected by frequent power interruptions.

In the Philippines, the average household experiences 28 electricity supply interruptions per year due to outages caused by plant breakdowns (Albay, 2025). In August 2025, the National Grid Corporation of the Philippines (NGCP) reported an unscheduled power interruption affecting Camiguin and specific parts of Misamis Oriental, which together house over sixty-thousand households (Philippine Statistics Authority, 2020). During such outages, reliance on diesel generators and other temporary energy sources underscores the inadequacy of existing solutions and the urgent need for reliable, affordable, and sustainable alternatives.

Recent efforts in solar-powered portable power supply systems show promise but still face limitations. Innovations have been made, such as Gozano et al. (2023) designing a solar-based portable power supply capable of charging small devices but with limited capacity and short battery life. Bhatti et al. (2024) developed a portable solar station with a 200 W output and load monitoring, while Ramly et al. (2019) created a 100 W portable solar power supply for emergencies that could last two days. Despite these advancements, existing systems still encounter issues related to energy capacity, battery lifespan, and limited support for higher power loads.

To address these gaps, this study proposes a solar-powered rental station with detachable power sources, powered by a solar panel for energy har-

vesting, a LiFePO_4 battery for reliable storage, and an inverter for AC power conversion. The system integrates an ESP32 microcontroller with IoT mechanisms to manage operations, while GPS and GSM modules provide real-time location tracking, communication, and monitoring of rented units, serving as additional security features of the system. A coin-slot mechanism with a solenoid lock ensures secure access, while the mobile application allows users to view available rental slots, check pricing based on the detachable power source's charge percentage, and manage user verification through SMS-based phone number confirmation. The application also stores the user's account information and available credits for future transactions. The compact and user-friendly design offers a sustainable solution for communities affected by frequent power interruptions, effectively bridging the gap in affordable and reliable energy access.

1.2 Statement of the Problem

Communities in various parts of the Philippines continue to face challenges in accessing stable electricity. In provinces such as Camiguin (served by CAMELCO), residents experience frequent and unscheduled power interruptions multiple times a week. These power outages cause inconvenience to households, affect business operations, and disrupt the productivity of local

communities. Consistent and stable access to electricity remains a persistent challenge, especially in rural and island areas.

Although alternatives such as off-grid systems, portable power banks, and small gasoline generators are available, they often prove costly, limited in capacity, or unsustainable in the long term. The use of fossil-fueled generators also raises safety and environmental concerns, which contradict current efforts toward cleaner and more sustainable energy sources.

Solar-powered systems have been introduced as alternatives, yet most existing systems can only supply limited power to small devices such as mobile phones, lamps, or AM/FM radios. Many of these systems are installed within college campuses and are not designed for community accessibility. In addition, the absence of security and monitoring mechanisms makes these systems prone to theft.

To address these challenges, this study aims to answer the following research questions:

1. How can a solar-powered rental station with detachable power sources be designed to provide a sustainable and user-friendly energy solution for communities experiencing frequent power interruptions?
2. How can the proposed system overcome existing limitations in solar-

powered portable power solutions, specifically lower energy load, limited battery capacity, restricted support for higher loads?

3. How can a secure coin-operated or mobile application-based mechanism be developed to regulate and monitor the rental and return of detachable power units?
4. How can GPS and GSM technologies be integrated to enhance tracking, monitoring, and communication of rented power sources?
5. To what extent can the developed prototype address issues of affordability, accessibility, and sustainability compared to existing backup power solutions?

1.3 Objectives of the Study

In view of the above stated problem, the following objectives are:

1.3.1 General Objectives

- To design and develop a solar-powered rental station with detachable power sources that are rented to serve as a sustainable and user-friendly solution for supplying electricity, specifically intended for communities experiencing frequent and unscheduled power interruptions.

1.3.2 Specific Objectives

- To design the system architecture for the power source that integrates photovoltaic panels, energy storage, and a power inverter, ensuring sufficient capacity for powering high-demand power appliances.
- To develop a functional prototype of the rental power station with a detachable power source, with a secure coin-operated or app-based access mechanism for rental use.
- To integrate GPS and GSM modules to support location tracking, communication, and monitoring of rented units.

1.4 Significance of the Study

This project benefits a diverse range of stakeholders including:

1.4.1 Community Residents: Community residents will benefit from continuous access to electricity through the detachable power source. This system ensures the community residents can use essential household appliances and stay connected during power interruptions.

1.4.2 Environment: This study contributes to environmental sustainability by promoting the use of solar energy, a renewable resource, instead of relying on fossil-fueled generators. The system reduces carbon emissions and pollution,

aligning with global sustainability goals.

1.4.3 Future Researchers: Future researchers will find this study helpful because it gives important information on how to create solar-powered portable charging systems with features like GPS tracking, coin/app payment, and monitoring.

1.4.4 SDG 7: Affordable and Clean Energy: This system provides support in giving people access to energy that is affordable, reliable, and environmentally friendly by offering a solar-powered rental station that provides communities, especially those often experiencing blackouts, with a clean and low-cost source of electricity.

1.4.5 Industry, Innovation, and Infrastructure: This helps achieve the goal of improving systems, creating new technologies, and building stronger infrastructure by using modern technologies, as well as solar energy, which helps in creating innovative solutions that give communities a reliable energy option during power interruptions.

1.5 Scope and Limitations

1.5.1 Scope

- Design a solar-powered portable power station with detachable power sources for communities in Camiguin and selected areas in Misamis Oriental, which face frequent power interruptions.
- Integrate photovoltaic panels, energy storage, and a power inverter to run high-demand power appliances.
- Implement coin-operated or app-based access for renting the power sources.
- Incorporate GPS and GSM modules for location tracking and communication, ensuring efficient monitoring of rented units.
- Weather-dependent operations: The system can operate during extreme weather, such as typhoons to protect users and equipment from potential damage

1.5.2 Limitations

- Geographic coverage limited to Camiguin and selected areas in Misamis Oriental.
- Cannot charge electric vehicles (e.g., Tesla) because required input is far higher than the system can supply.

- Deployment limited to suitable sites: Only for selected area that both (a) frequently experience power interruptions and (b) have adequate sunlight; performance is poor where solar access is obstructed (c) inoperable in areas without Internet connectivity
- Coins only: The coin-operated mechanism does not accept paper bills.
- No physical change: The system does not dispense change; any excess payment/remaining balance is credited to the user's mobile application account.

1.5.3 Definition of Terms

- Internet of Things (IoT) - A network of physical devices, such as sensors, appliances, and power sources, connected to the internet, enabling them to collect, send, and receive data for remote monitoring, control, and management.
- Solar Energy - Energy that is harnessed from sunlight using technologies such as photovoltaic (PV) panels, which convert sunlight into electricity, which is used to power the portable power sources.
- Fossil Fuel - Natural energy sources such as coal, oil, and natural gas, derived from the remains of ancient plants and animals, that are burned

to produce energy but contribute to environmental pollution and climate change due to the emission of greenhouse gases.

- Photovoltaic (PV) Panels - Solar panels that convert sunlight into electricity, serving as the primary source of power generation for solar-powered systems.
- Auxiliary load - The secondary electrical components that support the operation of the system, such as controllers, sensors, and communication modules. In this study, this includes the ESP32, GPS and GSM modules, coin-slot mechanism, solenoid lock, and other low-power electronics.
- Portable Power Station - Compact, mobile units that provide electrical power for charging devices or operating appliances, often powered by renewable sources like solar energy and designed to be easily transported or moved.
- Small-to-Medium Scale Appliances - Electrical devices that consume relatively low to moderate power, typically ranging from 10 watts up to around 800 watts during normal operation.
- High-Demand Power Appliances - Devices that require significantly high power, usually 800 watts to more than 3000 watts, and often cause large spikes in household electrical load.

- Renewable Energy - Energy derived from natural resources that are replenished on a human timescale, such as sunlight, wind, and geothermal heat, which are harnessed to produce electricity in an environmentally sustainable manner.
- Off-grid power system - A power system that operates independently from the main electricity grid, using renewable energy sources like solar or wind to provide electricity in areas without access to centralized power.
- Solar harvesting - The process of capturing sunlight using solar panels or other solar technologies and converting it into usable electrical energy, typically for storage in batteries or direct use.
- Power Interruption - A temporary loss or disruption of electrical power, often due to faults, maintenance, or other technical issues in the power grid, affecting the availability of electricity to households or businesses.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter highlights related projects or studies that offer valuable insights to the researchers, serving as a foundation for the development of the study.

2.1 Theoretical Background

This study is anchored on three major theories: the Rental System Theory, the Sustainable Energy and Portable Power Theory, and the Wireless Communication and Positioning Theory (GPS and GSM Modules). Together, these theories explain the technological and operational principles underlying the development and functionality of the Kuryenta system, a coin-operated detachable power rental station designed to provide accessible, sustainable, and flexible energy solutions to users.

2.1.1 Rental System Theory

The Rental System Theory provides the conceptual basis for automating resource access and transactions through integrated digital and physical mechanisms. Thakur (2021) describes how an online rental platform enables users to register, reserve, and manage bookings remotely, transforming manual

rental procedures into efficient, automated workflows. Applied to the Kuryenta system, this theory explains the combined use of a mobile application and on-station hardware (coin slot and solenoid mechanism) to control access to detachable power units. Users register and select an available slot via the application, then complete the physical payment at the station; the system computes the rental fee dynamically, based on the detachable unit's remaining battery percentage, and upon successful payment, actuates the solenoid to release the unit for a fixed rental duration. The integration of automated payment, real-time availability, and credit-storage mechanisms embodies core principles of rental system theory by increasing operational efficiency, ensuring transparent transaction management, and improving user convenience. These features align with Thakur's (2021) emphasis on system integration, automated database updates, and user-centred service delivery in modern rental platforms

2.1.2 Sustainable Energy and Portable Power Theory

The Sustainable Energy and Portable Power Theory supports the selection and design of renewable, off-grid power solutions and portable energy storage for resilient, community-oriented services. Recent implementations of coin-operated solar charging stations demonstrate how photovoltaic energy

harvesting coupled with integrated storage batteries can provide continuous, low-maintenance charging services with commercial and emergency applications (Catalan et al., 2023). Complementing this, Gozano et al. (2023) provide empirical evidence for the design and deployment of portable, modular battery packs charged by solar panels, including component sizing (PV, inverter, charge controller), battery configuration, and performance monitoring via Arduino data logging, to reliably serve remote communities. Applying these principles to Kuryenta justifies the use of detachable battery with an onboard inverter and supports design decisions such as panel and battery sizing, battery management (BMS/modularity), and performance validation protocols. Together, these studies reinforce key theoretical aspects of the Kuryenta system: energy sustainability (solar charging), adaptability (modular/detachable units and inverter capability for low-medium appliances), operational resilience (off-grid emergency use), and evidence-based system sizing and monitoring for effective, user-centered portable power delivery.

2.1.3 Wireless Communication and Positioning Theory (GPS and GSM Modules)

The Wireless Communication and Positioning Theory serves as the foundation for the integration of the Global Positioning System (GPS) and

Global System for Mobile Communication (GSM) modules within the proposed system. This theory emphasizes the use of satellite-based navigation and mobile communication technologies to enable real-time tracking, monitoring, and data transmission over long distances. According to San Hlaing, Naing, and San Naing (2019), the integration of GPS and GSM enables efficient and continuous monitoring of mobile assets by acquiring geographic coordinates and transmitting them via wireless networks to designated users or databases.

In the context of this study, the theory supports the functionality of tracking and monitoring the location of detachable power sources within the rental system. The GPS module gathers real-time location data, while the GSM module transmits these coordinates to the mobile application and administrative system. Through this process, users and administrators are able to monitor the deployed power units, ensuring accountability, operational transparency, and user security. This theoretical foundation highlights the importance of wireless connectivity and geolocation in developing intelligent systems that enhance mobility, accessibility, and user convenience in portable power management.

2.1.4 Integration of Theories

All three theories collectively form the foundation of this study. The Rental System Theory provides the framework for automating the process of accessing, reserving, and paying for detachable power units through a combination of digital applications and physical mechanisms, ensuring efficient and user-centered rental transactions. The Sustainable Energy and Portable Power Theory establishes the system's grounding on renewable and off-grid energy utilization, promoting sustainability and operational resilience through solar-powered charging and modular battery storage. Meanwhile, the Wireless Communication and Positioning Theory supports the integration of GPS and GSM modules that enable real-time monitoring, data transmission, and location tracking of the detachable power sources. Together, these theories support the development of KURYENTA: Solar-Powered Rental Station with Detachable Power Sources, a coin-operated, GPS-enabled, and solar-powered system designed to deliver accessible and sustainable portable energy solutions to users.

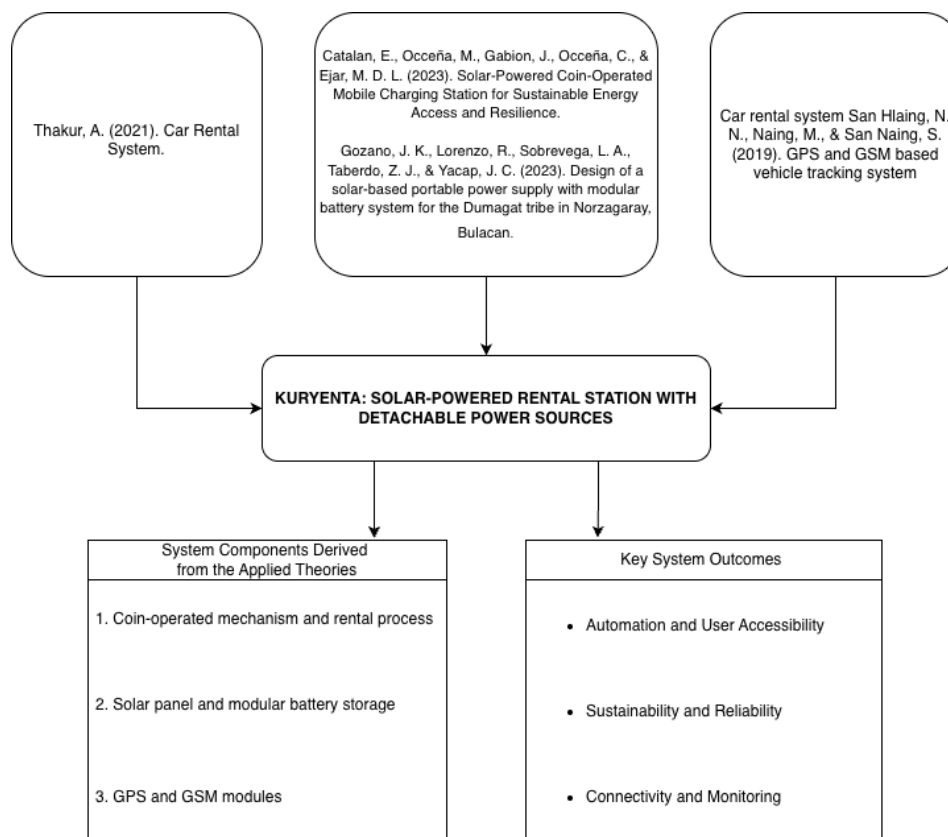
Figure 2.1*Theoretical framework*

Figure 2.1 illustrates the Theoretical Framework of the study, showing how the Rental System Theory (Thakur, 2021), Sustainable Energy and Portable Power Theory (Catalan et al., 2023; Gazano et al., 2023), and Wireless Communication and Positioning Theory (San Haing et al., 2019) collectively form the conceptual basis of KURYENTA: Solar-Powered Rental Station with Detachable Power Sources.

These theories complement one another in explaining the system's core components, automation, sustainability, and connectivity, thereby providing a coherent theoretical foundation for its design and development.

2.2 Conceptual Framework

The conceptual framework of the study illustrates the overall structure of how the solar-powered rental station functions and how each part of the system contributes to its operation. It introduces the importance of the Input-Process-Output (IPO) model, which outlines how raw inputs are transformed through system processes into useful outputs. The framework also presents the Independent Variables, which include the user actions and environmental factors that influence system performance. Finally, it identifies the Dependent Variables, which represent the resulting outputs and behaviors of the system based on those inputs and processes.

Figure 2.2

IPO Model

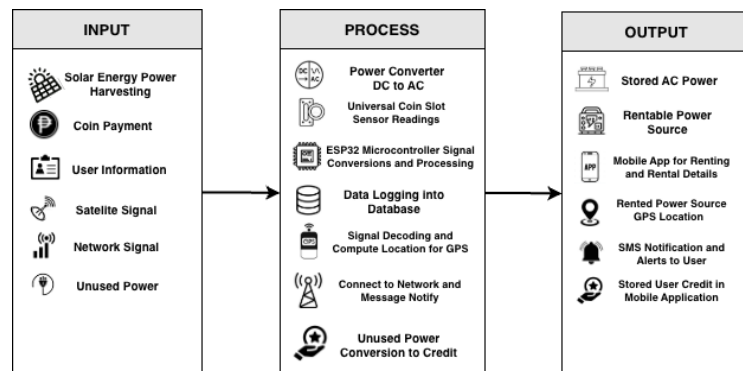


Figure 2.2 illustrates how the solar-powered rental station transforms raw inputs into useful results. Inputs include solar energy, coin payments, user data, satellite and network signals, and unused power. These pass through processes like power conversion, sensor detection, microcontroller computation, database logging, GPS decoding, network communication, and credit calculation. The system then produces outputs such as stored AC power, rentable power sources, app rental details, GPS-tracked units, SMS alerts, and updated user credits.

a. Variables

- **Independent Variables (IV)**

Coin, Credit, Distance, Signal, Unused Power

- **Dependent Variables (DV)**

Power, Location, Notification, Credit

b. Formal Definition of Variables

Independent Variables

- *Coin* – Serves as the physical input medium that activates the system's rental process through coin insertion, enabling users to access the detachable power source.

- *Credit* – The amount of digital money or balance that a user has in their account (through the mobile app). The user can use this balance to pay for renting the detachable power source.
- *Distance* – The spatial measurement between the rental station and the portable power source as tracked by the GPS module.
- *Signal* – The communication strength and reliability between the system’s GPS and GSM modules. It affects the accuracy of data transmission, notifications, and location tracking.
- *Unused Power* – Refers to the remaining electrical energy stored in the detachable power source after it has been rented and used by the user. This remaining power is measured based on the battery’s remaining capacity percentage when returned. The system uses this value to calculate the amount of credit or reward that will be given to the user through the mobile application

Dependent Variables

- *Power* – The amount of electrical energy stored and supplied by the solar-powered battery system. It determines the duration and capacity of electricity available for users during rental operations.
- *Location* – The position of the rented portable power source as mon-

itored by the GPS module. It indicates where the unit is located during operation.

- *Notification* – Refers to the alerts sent to the user via the GSM module. It notifies the user one hour before the allotted 12-hour usage period ends and informs them to return the item if the time exceeds the 12-hour limit. Notifications are then sent at 10-minute intervals until the item is returned.
- *Credit* – Refers to the digital value or reward points that the user earns in the mobile app after returning the rented detachable power source. The credit amount is calculated based on how much unused power remains in the device upon return.

c. Relationship of Variables

The independent variables directly influence the dependent variables.

- *Coin $\rightarrow$ Power*

Inserting a coin activates the rental process, allowing the user to access the detachable power source. The available power supplied by the system depends on a valid coin being inserted,

- *Distance $\rightarrow$ Location*

The measured distance of the detachable power source from ref-

erence points or the base station affects the GPS-based location tracking. The reported location depends on this spatial relationship, as distance influences the precision of GPS signals.

- *Signal $\rightarrow$ Location*

The strength of the GPS signal affects the accuracy and reliability of location tracking.

- *Signal $\rightarrow$ Notification*

The GSM signal strength determines how effectively the system can send alerts and messages to the user.

- *Unused Power $\rightarrow$ Credit*

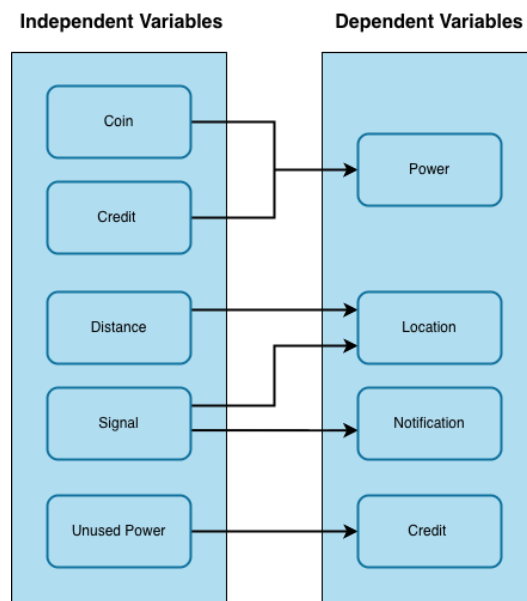
The amount of unused power remaining in the portable battery after a rental session determines the credit returned to the user through the application.

- *Credit $\rightarrow$ Power*

The credit available in a user's account can be used to pay for the rental, which then activates the detachable power source.

2.2.1 Schematic Diagram

Figure 2.3 below presents the independent and dependent variables, along with their relationship.

Figure 2.3*Variable relation*

2.3 Review of Related Studies

This part highlights related projects or studies that offer valuable insights to the researchers, serving as a foundation for the development of the study.

2.3.1 Design of a Solar-Based Portable Power Supply with Modular Battery System for the Dumagat Tribe in Norzagaray, Bulacan

Access to reliable and sustainable energy is a critical issue in many rural and indigenous communities. One such community is the Dumagat tribe in

Norzagaray, Bulacan, where there is a notable lack of access to electricity and public utilities despite the usage of electronic devices like phones. Addressing this issue requires innovative solutions that not only meet their energy demands but also ensure sustainability and accessibility.

The study of Gozano et al. (2023) developed a solar-powered portable power supply for the Dumagat tribe, integrating a solar panel, modular battery pack, and inverter to meet basic energy needs for low-power devices. It is designed to supply low-energy loads such as mobile phone charging, portable lighting, and small auxiliary devices like a small AM/FM radio. The system was tested using an Arduino data logger to measure charging and discharging rates, demonstrating effective energy delivery for community use. However, the study identified the need for higher-capacity batteries and more efficient solar panels to improve sustained energy supply and enhance system performance. Future improvements could focus on optimizing battery storage and panel efficiency to ensure longer operational hours and accommodate higher energy demands.

2.3.2 Solar-Powered Coin-Operated Mobile Charging Station for Sustainable Energy Access and Resilience

The increasing demand for sustainable and accessible energy has driven innovations that utilize renewable sources to meet electricity needs in both urban and rural areas. The study by Catalan et al. (2023) aimed to develop a solar-powered, coin-operated mobile charging station designed to provide continuous off-grid power using photovoltaic (PV) technology. Employing a developmental research approach, the researchers designed, fabricated, and tested a prototype equipped with solar panels and an integrated storage battery system capable of charging multiple mobile devices for both commercial and emergency purposes. The results demonstrated that the charging station effectively powered various mobile gadget models with no compatibility issues, maintained stable operation even under limited sunlight, and offered cost efficiency through its low maintenance and sustainable design. Furthermore, the system contributed to reducing carbon emissions and supported the green technology initiatives of Guimaras State University. A key gap is the integration of secure access mechanisms and broader community-focused applications, beyond mobile device charging. It was concluded that the solar-powered charging station is a practical and eco-friendly innovation that promotes energy resilience in communities affected by power outages. The study

recommended installing such systems in strategic locations for communal purpose, especially in far-flung and island barangays, integrating security features to deter users' delinquent acts in the system, and conducting further research to enhance its technological design and promote the widespread adoption of renewable energy solutions.

2.3.3 Portable Solar-Station with Integrated Battery Management and Load Monitoring System

Abdur et al. (2024) describe the development of a portable solar-powered station designed for disaster response, offering a 200W output using a 2x2 array of 50W solar panels. The system integrates a battery management system (BMS) and load monitoring mechanism (LMM), providing power for essential needs such as lighting, mobile device charging, and medical equipment in emergency situations. Replacing diesel generators reduces environmental pollution and noise, offering a sustainable, clean energy solution in areas with limited infrastructure. Its compact design allows easy transportation to disaster zones, making it an effective tool for providing reliable power where it's needed most.

The system reduced reliance on diesel generators, offering a quieter and environmentally friendly solution. While the design is compact and portable,

there is room to further optimize portability and energy efficiency for rapid deployment in emergency situations. Future enhancements could include lighter materials and improved battery performance to support longer operational periods.

2.3.4 Multiport Universal Solar Power Bank

Altelmessani et al. (2024) introduce the concept of a Multiport Universal Solar Power Bank, designed to harness solar energy for a portable power supply. This device aims to address critical needs in emergency situations and recreational activities like camping, especially in remote areas where access to electricity is limited. Equipped with solar panels for efficient energy absorption, the power bank offers both DC and AC outputs, making it versatile enough to charge a wide range of electronic devices. The system can act as an AC power source with a maximum output of 400W and includes a battery capacity of about 20,000mAh, making it capable of powering various small appliances and electronic devices. It also incorporates safety features to protect against overcharging, overheating, and other risks, ensuring both user safety and device longevity. However, long-term durability under diverse environmental conditions remains a gap, which could affect performance in field applications. Recommendations include strengthening protective features and

testing the system under varied operational scenarios to ensure reliability.

2.3.5 Portable Power Supply Design with 100 Watt Capacity

Zakri et al. (2021) developed a portable solar power supply design with a 100W capacity, aimed at providing sustainable energy in areas with limited electricity access. The system utilizes solar cells and a transformer to store energy in batteries with capacities of 20Ah, 60Ah, and 100Ah. The solar-powered generator can charge devices like fan, lamps, laptops, televisions, supporting electrical loads below 100 watts. The system includes a solar charge controller, safety features, and an LCD for monitoring voltage, offering practical and environmentally friendly solutions for rural communities. A limitation is that efficiency depends on battery capacity and sunlight conditions, affecting continuous power availability. Future improvements could focus on optimizing battery sizing and charging efficiency for consistent energy delivery.

2.3.6 Design and Development of Portable StandAlone Solar Power Generator

Prathiba et al. (2020) developed a portable, standalone solar power generator designed to replace diesel generators with a sustainable and eco-friendly solution. The system integrates a solar panel, a battery, a bidirectional buck-boost converter, and an inverter, all supported by a Maximum

Power Point Tracking (MPPT) algorithm for optimal efficiency. The generator provides a green energy source to meet load requirements and stores excess energy in a battery for use when solar energy is unavailable. The bidirectional converter enhances battery charging efficiency and ensures regulated DC voltage output, while the MPPT algorithm maximizes power extraction from the solar panel to improve overall system performance.

The system was mechanically designed to allow PV panels to be mounted on the roofs of vehicles, enabling power generation above 1 kW. To achieve this 1 kW output, the storage system use lithium-ion batteries due to their high energy density, which provides better storage capacity and efficiency. The portable solar generator also utilizes a bidirectional converter and MPPT to efficiently charge and discharge a 12V lead-acid battery. While the design is effective and compact, the study notes potential for further improvement in energy extraction and storage. Recommended enhancements include advanced MPPT algorithms and converter optimization to increase system performance and reliability, particularly in remote locations.

2.3.7 A solar-powered multi-functional portable charging device (SPMFPCD) with internet-of-things (IoT)-based real-time monitoring- An innovative scheme towards energy access and management

Rehman et al. (2024) propose a solar-powered multi-functional portable charging device (SPMFPCD) with IoT-based real-time monitoring, designed to address the growing need for reliable and versatile energy solutions across various sectors, including transportation, communication, and emergency services. The device integrates a highly efficient solar panel, a charge controller, sensors, and an IoT module for real-time monitoring of power parameters. The study emphasizes the significance of integrating advanced technologies, such as IoT-driven battery energy storage system (BESS) health monitoring, to optimize the performance and lifespan of the system.

The system integrates solar panels, a charge controller, sensors, and IoT modules to track energy usage, battery health, and system status, ensuring efficient and reliable operation. A key gap is assessing the scalability and feasibility of widespread deployment for community-based energy solutions. The study recommends further testing and expansion to evaluate operational costs, performance, and adaptability in different environments while maintaining sustainable and accessible energy provision.

2.3.8 Renewable Energy from Solar Panels: A Study of Photovoltaic Physics and Environmental Benefits

Jaiswal (2023) provides an in-depth analysis of solar energy's growing role in the global energy transition, focusing on its environmental, economic, and technological advantages. The study highlights that global solar photovoltaic (PV) capacity reached about 1,059 gigawatts by 2021, reflecting rapid adoption and its significant contribution to reducing greenhouse gas emissions. Technological advancements, such as bifacial and perovskite solar cells, have increased the efficiency and affordability of solar power, making it more accessible. The research also stresses the importance of supportive policies and regulatory frameworks in promoting solar energy deployment.

The environmental benefits of solar power are significant, with Jaiswal (2023) noting that solar energy could reduce up to 80% of greenhouse gas emissions by 2050. Additionally, the integration of solar energy with energy storage systems is essential for improving reliability and addressing challenges related to intermittency. The study concludes that solar energy plays a crucial role in achieving sustainable development, reducing climate change impacts, and driving economic growth. With ongoing technological advancements and comprehensive policies that encourage investment in renewable energy, solar power is set to be a key component of the future energy system.

2.3.9 Power Consumption of Household Appliances

Power consumption patterns in households are significantly influenced by the frequency and duration of appliance usage. Pulvera (2021) found that household appliances such as televisions, electric fans, and refrigerators are among the most frequently used devices, contributing considerably to total electricity consumption. The results revealed that the television set operates for an average of forty-four (44) hours per week, followed by the electric fan with fifty-one (51) hours, and the refrigerator with one hundred five (105) hours of usage weekly. These appliances are commonly prioritized for their essential roles in providing comfort, entertainment, and food preservation within the household.

Pulvera (2021) further emphasized that electricity usage is affected by several factors, including low voltage supply, appliance wattage, power interruptions, standby power, and user behavior. Among these, user awareness and proper energy management play a crucial role in reducing unnecessary power consumption. The continuous use of high-demand appliances such as electric fans and refrigerators, especially in tropical climates like the Philippines, highlights the dependence of households on these devices for maintaining comfort during hot weather and ensuring food preservation. Consequently, understanding how frequently appliances are used helps identify energy-saving

opportunities, promote consumer awareness, and develop better methods to reduce energy costs, such as testing new systems or designing technologies with sustainability in mind.

2.3.10 Electricity Distribution and Supply Authority

Battery selection plays a critical role in optimizing the performance and sustainability of photovoltaic (PV) systems. According to the Electricity Distribution and Supply Authority (EDSA, 2024) under the Government of Sierra Leone, three major battery options are typically considered in solar power applications—Flooded Lead Acid, Sealed Lead Acid (SLA), and Lithium Iron Phosphate (LiFePO_4). The study highlighted that LiFePO_4 batteries are preferred for the Regional Emergency Solar Power Intervention (RESPITE) Project due to their superior efficiency, longevity, and environmental advantages.

LiFePO_4 batteries are generally more energy-efficient, allowing for greater energy utilization and reduced losses during charging and discharging. In contrast, SLA batteries exhibit lower efficiency, which can result in higher energy wastage. Additionally, LiFePO_4 batteries support a higher depth of discharge (DoD), enabling deeper energy use without significantly affecting their lifespan. This characteristic makes them more suitable for daily solar energy cy-

cling compared to SLA batteries, which tend to degrade faster under similar conditions.

Another major advantage of LiFePO_4 technology is its longer cycle life. The EDSA (2024) report notes that LiFePO_4 batteries can endure more charge-discharge cycles before their performance diminishes, offering better long-term reliability. They also possess higher energy density, meaning they can store more energy in a smaller physical footprint—an important factor when space constraints exist in solar installations.

In terms of maintenance, LiFePO_4 batteries are maintenance-free, unlike SLA batteries which often require periodic electrolyte checks and refilling. While LiFePO_4 batteries have a higher upfront cost, they compensate with lower operational and maintenance expenses throughout their lifespan. From an environmental perspective, LiFePO_4 batteries are also less harmful and easier to dispose of or recycle compared to lead-acid types, which generate toxic emissions during recycling.

2.3.11 Assessing the Impact of Power Outages on Appliances of Farmers and Fisherfolks in Selected Barangays of Cawayan, Masbate, Philippines: Basis for a Proposed Extension Program

Frequent power interruptions have long been a challenge in rural communities, particularly among farmers and fisherfolk who depend on electricity for household and livelihood activities. The study aimed to determine the impact of frequent power interruptions on the appliances and economic well-being of residents in selected barangays of Cawayan, Masbate. Employing a descriptive research design, the researchers gathered data from 266 respondents through paper surveys and face-to-face interviews to assess the frequency, duration, and effects of power interruptions. Findings revealed that almost all respondents experienced power interruptions lasting three to four hours, leading to increased electricity consumption and higher bills. Refrigerators and televisions were the most power-consuming appliances, and there was significant damage to appliances, especially bulbs, as well as disruptions to income-generating activities. The study results showed that many respondents had an annual income of less than P18,200, which was considered low and may have resulted in difficulty in paying high bills brought by power outages. All respondents relied on the power grid as their source of electric-

ity, and power interruptions were a common occurrence. The data revealed that 97.7% of respondents experienced power interruptions, with 51.1% experiencing 3-4 hours of interruption. Almost all respondents claimed that power interruption increased their electric consumption and bill, and 56% were not satisfied with their electric bill when there was a power interruption. It was concluded that unreliable electricity supply and lack of maintenance significantly affect low-income households that rely solely on the power grid as their source of electricity. The study recommended It is strengthening local infrastructure and integrating sustainable energy solutions be prioritized to support low-income households and livelihoods dependent on electricity. Additionally, there is potential to increase the adoption of solar energy, given the positive perception of its benefits among residents. Implementing projects like Electrical and Electronic Designs using Renewable Energy across barangays can enable communities to harness free, sustainable energy from the environment, reduce reliance on the traditional power grid, mitigate environmental impacts, and enhance resilience to climate-related challenges.

2.3.12 Synthesis

The reviewed literature highlights the development of solar-powered portable energy systems aimed at providing reliable power to off-grid and re-

mote communities. Earlier studies, such as those by Gozano et al. (2023), Zakri et al. (2021), and Prathiba et al. (2020), integrated solar panels, batteries such as 12V lead-acid batteries, charge controllers, and inverters to support basic appliances and emergency needs. However, many of these systems had limited energy storage capacity or relied on conventional batteries, restricting continuous power supply. According to EDSA (2024) and Zakri et al. (2021), LiFePO_4 batteries offer advantages including higher cycle life (over 2,000 cycles), greater depth of discharge (DoD), better efficiency, lower maintenance, and environmental friendliness, making them suitable for systems with frequent charge–discharge cycles.

More recent studies by Abdur et al. (2024), Altelmessani et al. (2024), and Rehman et al. (2024) incorporated advanced features such as battery management, load monitoring, and IoT-based tracking to enhance energy efficiency and system reliability. Nevertheless, gaps remain in providing secure, user-controlled access to stored energy, limiting the practical deployment of these systems in community and household contexts. In addition, most designs focus on individual appliances or emergency loads, leaving opportunities for broader community-level applications, such as powering shared spaces in far-flung or island barangays and strategically placing systems for maximum communal benefit (Catalan et al., 2023).

Several studies also highlight cost reduction and operational efficiency as key considerations. Implementing solar energy systems with high-capacity batteries, integrated monitoring apps, and pay-per-use mechanisms can reduce electricity costs, encourage responsible consumption, and ensure equitable access. Security features are necessary to deter misuse and protect system integrity, especially in communal settings. Furthermore, such systems should be designed to accommodate higher energy demands (Gozano et al., 2023) and enhance the technological design and widespread adoption of renewable energy solutions (Catalan et al., 2023).

To address these gaps, recent approaches propose combining high-capacity LiFePO_4 batteries with coin-slot or pay-per-use mechanisms for secure access, along with mobile app monitoring through ESP32, GPS, and GSM modules to provide updates on energy consumption and system status (Abdur et al., 2024; Rehman et al., 2024). Integrating a 2x2 solar panel array, MPPT charge controller, and AC inverter enables the system to power higher-demand appliances, including air conditioning units and washing machines, which are commonly used in Filipino households (Pulvera, 2021). This configuration supports sustainable, accessible, and efficient energy management while reducing operational costs, encouraging responsible consumption, and promoting wider adoption of renewable energy systems in remote communities (Catalan et al.,

2023; Gozano et al., 2023).

Overall, the system’s sustainable design—characterized by high-capacity LiFePO_4 batteries, secure rental features, and monitoring—provides a comprehensive solution that builds on existing research while filling the gap in existing solar energy systems for communities that frequently face power interruptions.

CHAPTER 3

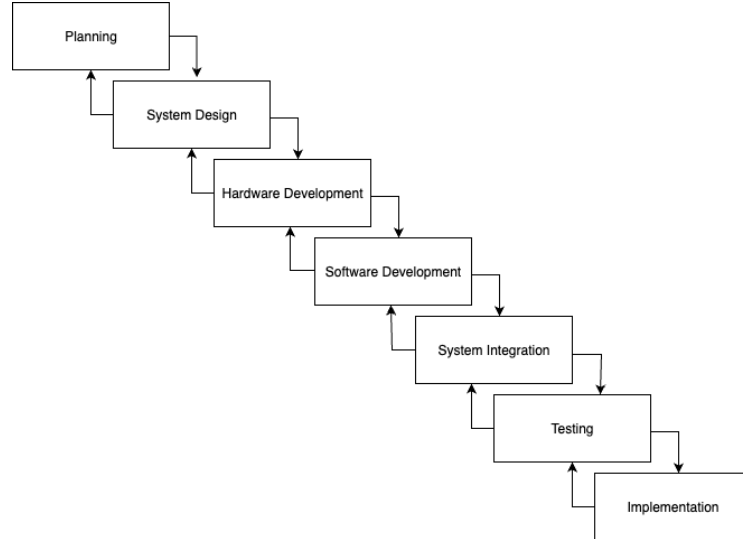
METHODOLOGY

In this chapter, the researchers outline the methodology employed to conduct the study, providing a comprehensive overview of the research design, technical design workflow, and analytical procedures.

3.1 Research Design and Procedure

Figure 3.1

The Waterfall Model



In Figure 3.1, the system creation process is illustrated using a modified waterfall model. This sequential approach consists of several key stages, beginning with planning and followed by system design, hardware development,

software development, system integration, testing, and finally, implementation.

The waterfall model is traditionally a linear development framework where each phase must be completed before the next begins. Its modified version, used in this study, maintains this structured flow while allowing limited feedback or adjustments in earlier stages. This makes it suitable for hardware–software projects, where clear documentation, defined requirements, and step-by-step validation are essential. The model ensures that system components are carefully planned, designed, and tested before full integration, reducing the likelihood of major errors during later phases.

This procedural framework guides the researchers in achieving the study’s intended objectives by ensuring that each step is completed before moving forward. It also supports early corrections, which helps minimize development costs and ensures that the hardware and software components are aligned with system requirements.

3.1.1 Planning

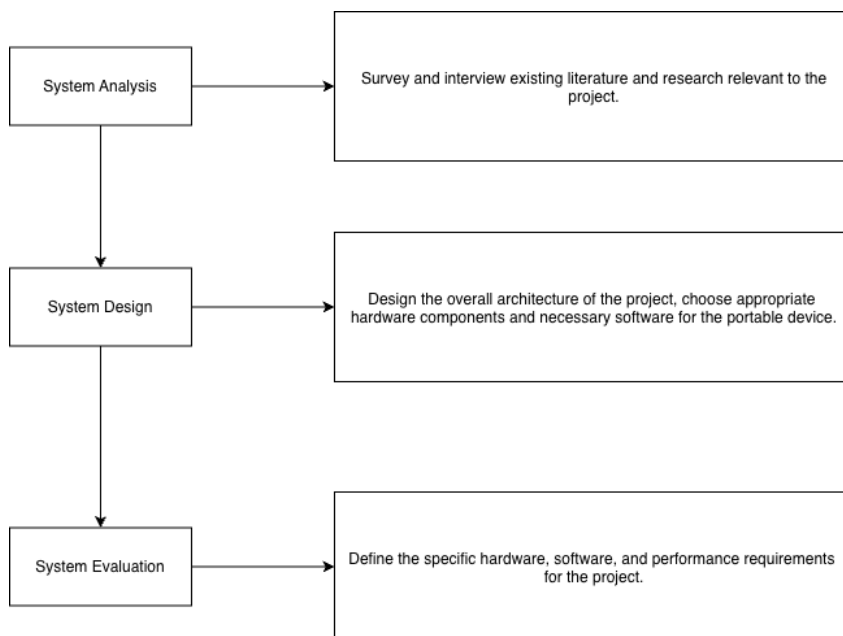
The planning phase serves as the first and foundational stage of the Waterfall Model. In this phase, the overall direction of the project is established by identifying the problem, defining the system requirements, and determining the objectives and scope of the study. This stage also involves gathering

relevant information, assessing user needs, and outlining the resources, constraints, and timelines essential for system development.

Through careful planning, the researchers ensure that the succeeding phases—system design, hardware and software development, integration, testing, and implementation—are built on a clear and structured framework. By employing a systematic methodology, the study is able to conduct an in-depth examination of the research domain and prepare the necessary groundwork for a successful development process.

Figure 3.2

System Development Life Cycle



The System Development Life Cycle (SDLC) for this study is illus-

trated in the figure, showing the sequential stages of system development. The process begins with System Analysis, where researchers conduct surveys, interviews, and review existing literature and studies relevant to the project. This phase ensures a thorough understanding of the system requirements, user needs, and potential constraints.

Following this, the System Design phase involves planning the overall system architecture. During this stage, appropriate hardware components, such as solar panels, LiFePO_4 batteries, microcontrollers, and monitoring modules—are selected, along with the necessary software to control, monitor, and manage the portable energy device. The design phase ensures that both hardware and software are compatible, efficient, and capable of fulfilling the project objectives.

The System Evaluation phase focuses on defining specific performance criteria and testing the system's components. This includes evaluating hardware specifications, software functionality, and overall system efficiency to ensure that the portable solar-powered energy system operates as intended. Through this phase, any shortcomings can be identified and addressed before full implementation, guaranteeing a reliable and functional final product.

Budget Planning

The development of the solar-powered rental station with detachable power sources, intended as a solution for supplying electricity to communities experiencing frequent and unscheduled power interruptions, incurred a total cost of P27,140.00. This amount covered the materials used in the study, the construction of the device, and the data collection and testing conducted throughout the research.

Research Locale

Figure 3.3

Map of Camiguin



The research will be conducted in specific area of Camiguin, Philippines that is known for frequent and unscheduled power interruptions. This area is served by CAMELCO (Camiguin Electric Cooperative), which experience regular power outages, particularly during peak hours. This issue disrupts daily activities, especially in community centers, educational institutions, and local businesses that rely on electricity for mobile devices, lighting, and appliances.

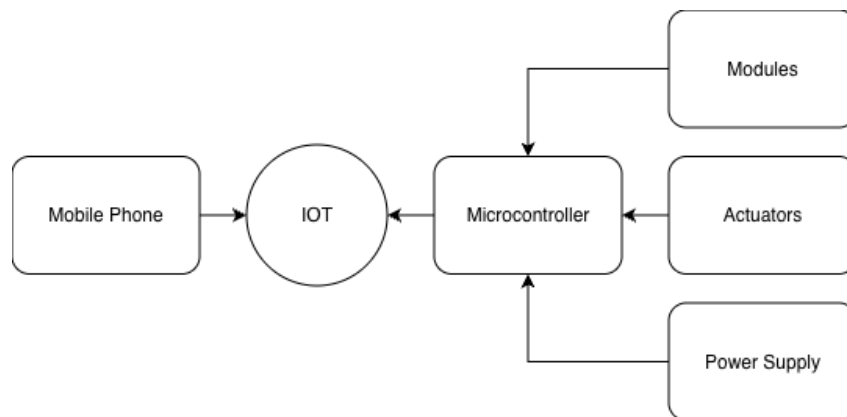
3.1.2 System Design

This section outlines the system design. It

3.1.3 Hardware Development

Figure 3.4

Hardware Development Diagram



Material Specifications

This section outlines the criteria and specifications for the materials and components used in the final device prototype. Each component was chosen based on specific criteria to ensure —

The following table outlines the key considerations for choosing hardware components for a system. It details the specific features, specifications, and desired functionalities for each component.

Table 1: Criteria for Selecting Hardware Materials

Component	Feature	Specifications
Microcontroller	Connectivity	Wi-Fi, Bluetooth
	Clock Speed	Up to 240 MHz
	RAM	520 KB SRAM
	Flash Memory	4 MB
	Power Supply	3.3V
Solar Panel	Power	410W
	Voltage	~36V
	Current	~11.4A
	Efficiency	~20%
	Cell Type	Monocrystalline
Battery	Voltage Rating	12V
	Battery Capacity	50-100 Ah
	Output Energy	0.6-1.2 kWh
	Cycle Life	2000+ cycles
	Type	LiFePO ₄
Charge Controller	Input Voltage	12V / 24V auto
	Maximum Charge Current	30A
	Max PV Power	500W
	Efficiency	>95%
Inverter	Input Voltage	12V DC

Component	Feature	Specifications
	Output Voltage	220V AC
	Continuous Power	1000W
	Peak Power	2000W
	Output Waveform	Pure Sine Wave
Battery Charger	Output Voltage	14.6V
	Output Current	20A
	Power	292W
Coin Slot Facade	Acceptable Coins	Universal
	Operating Voltage	12V DC
DC Breaker	Rated Current	40-80A DC
	Voltage Rating	12-24V DC
Flexible Silicone Wire	Gauge	4-10 AWG
	Voltage Rating	600V
	Current Capacity	30-70A
	Length	0.5-1 m
GSM Module	Operating Voltage	5V DC
	Network	2G / 3G GSM
GPS Module	Operating Voltage	3.3V-5V DC
	Accuracy	± 2.5 m
	Update Rate	1 Hz
Relay	Coil Voltage	12V DC
	Contact Rating	10A / 250V AC
Solenoid Lock	Voltage	12V DC
	Current	1A
	Holding Force	200 N
AC Breaker	Rated Current	4A AC
	Voltage Rating	220V AC
LCD Display	Display Type	16X2 Characters
	Interface	I2C
	Operating Voltage	5V DC
Router	Input Voltage	5V DC
	Wireless Standard	Wi-Fi 802.11 b/g/n

Table 2: Selection of Hardware Components

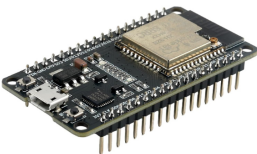
Hardware Materials	Description
 ESP32	The researchers selected the ESP32 microcontroller as the central processing unit of the system. Its built-in Wi-Fi and Bluetooth modules, low power consumption, and compact size make it suitable for IoT-based applications. The ESP32 efficiently manages all sensors, relays, and peripheral devices, providing reliable performance for the rental power station.
 Monocrystalline Solar Panel	A monocrystalline solar panel was chosen for its high efficiency and durability in converting sunlight to electrical energy. This ensures sufficient energy production for battery charging, even under low-light conditions. Its compact and sturdy design makes it ideal for the portable solar power system used in the project.
 LiFePO₄ Battery	The LiFePO₄ battery was selected for its high energy density, long cycle life, and safety advantages over other lithium-ion batteries. It provides a stable and reliable power supply for the system, supporting continuous operation of the microcontroller and connected devices. Its capacity was matched to the energy requirements of the portable rental unit.
Continued on next page	

Table 2 – continued from previous page

Hardware Materials	Description
 MPPT Solar Charge Controller	An MPPT (Maximum Power Point Tracking) solar charge controller was incorporated to optimize energy harvest from the solar panel. It ensures efficient and safe charging of the battery by adjusting voltage and current according to the panel's output, extending battery life and maintaining system reliability.
 Pure Sine Wave Inverter	The pure sine wave inverter was chosen to convert DC battery power to AC power for appliances. It provides clean and stable electricity, preventing potential damage to devices from voltage fluctuations, and ensures reliable energy delivery to users of the rental power station.
 LiFePO₄ Battery Charger	A dedicated LiFePO₄ battery charger was used to safely recharge the battery when connected to an external power source. This ensures efficient energy replenishment while protecting the battery from overcharging, enhancing the longevity and reliability of the portable system.
Continued on next page	

Table 2 – continued from previous page

Hardware Materials	Description
 Coinslot Multi Coin Selector	The coinslot multi coin selector was selected as a payment input device for the rental system. It allows users to securely deposit coins of different denominations, ensuring accurate transaction processing. Its reliability and ease of integration make it suitable for a user-friendly payment system.
 DC Breaker	A DC breaker was included to protect the system from overcurrent or short circuits in the DC circuit. Its presence ensures safety for both the users and the system components, minimizing the risk of damage to the battery, solar panel, or other hardware elements.
 Flexible Silicone Wire	Flexible silicone wires were chosen for their durability, high temperature tolerance, and flexibility. They ensure safe and efficient electrical connections between components, contributing to the overall reliability and portability of the hardware system.
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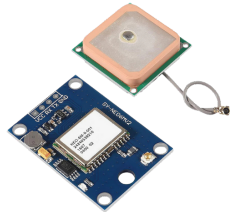
Hardware Materials	Description
 GSM module	The GSM module was selected to enable remote communication between the rental system and the server. It facilitates sending SMS alerts or notifications regarding rental status, battery level, and system operations, ensuring timely monitoring and management.
 GPS Module	A GPS module was incorporated to track the location of the portable power units. This allows for real-time monitoring and management of the units, enhancing security and operational efficiency in the rental system.
 Relay	Relays were used to control high-voltage components using low-voltage signals from the microcontroller. Their selection ensures safe and efficient switching of power to various devices, protecting the system from electrical overloads.
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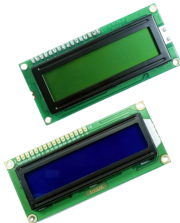
Hardware Materials	Description
 Solenoid Lock	The solenoid lock was chosen to secure the portable power unit. It allows controlled access based on successful rental authorization, ensuring security while maintaining ease of operation for users.
 AC breaker	An AC breaker was included to safeguard the AC output circuit. It prevents potential damage from over-current or short circuits when appliances are connected, ensuring both user safety and system protection.
 16X2 LCD Module with I2C	The 16x2 LCD module with I2C interface was selected to provide clear visual feedback to users. It displays relevant information such as rental instructions, system status, and battery level. The I2C interface reduces wiring complexity and enhances system integration.
Continued on next page	

Table 2 – continued from previous page

Hardware Materials	Description
 Wi-Fi Router	A Wi-Fi router was used to provide wireless connectivity between the system and the mobile application. Its inclusion ensures stable communication for data transfer, remote monitoring, and user interactions, supporting the IoT-based functionality of the rental power system.

Circuit Diagram

Figure 3.5

Circuit Diagram of Rental Station

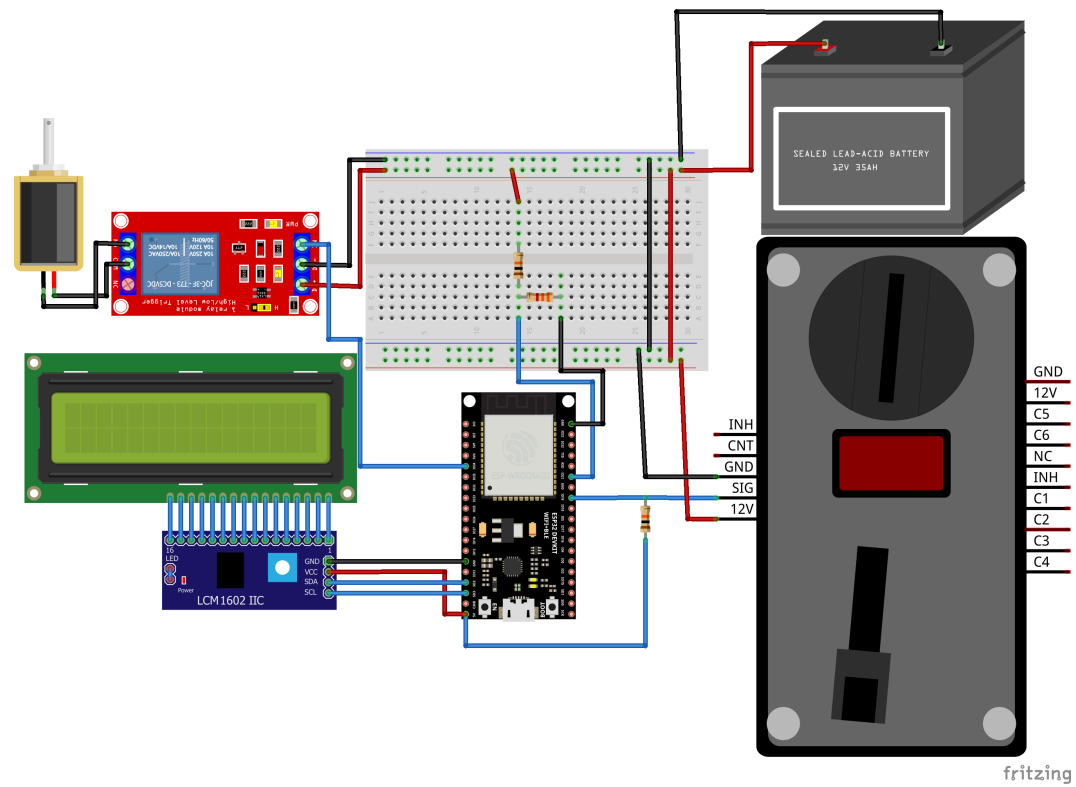
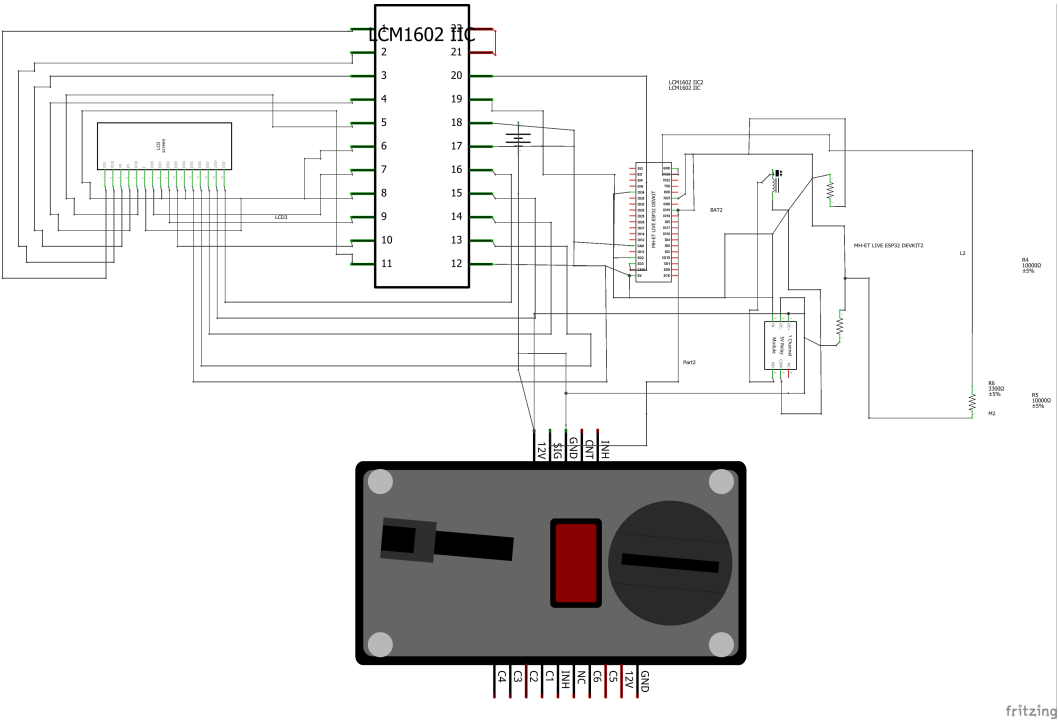


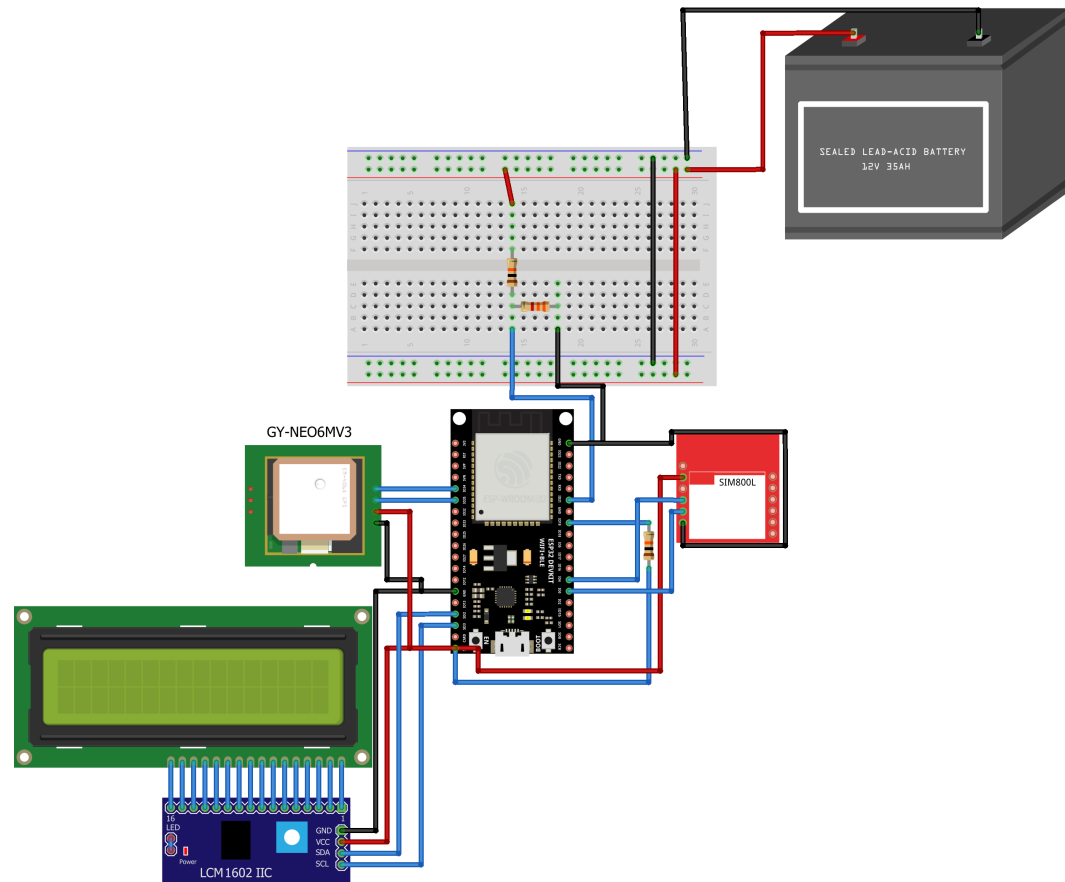
Figure 3.6
Circuit Diagram of Detachable Power Sources



Schematic Layout

Figure 3.7

Schematic Diagram of Rental Station

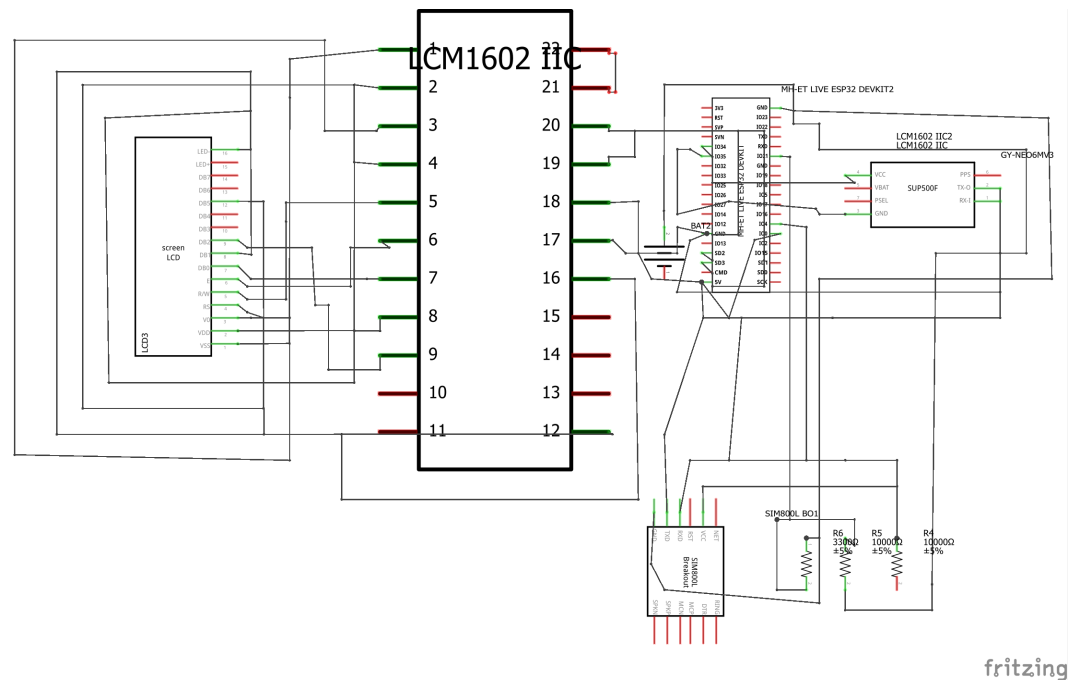


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Schematic Layout

Figure 3.8

Schematic Diagram of Detachable Power Sources

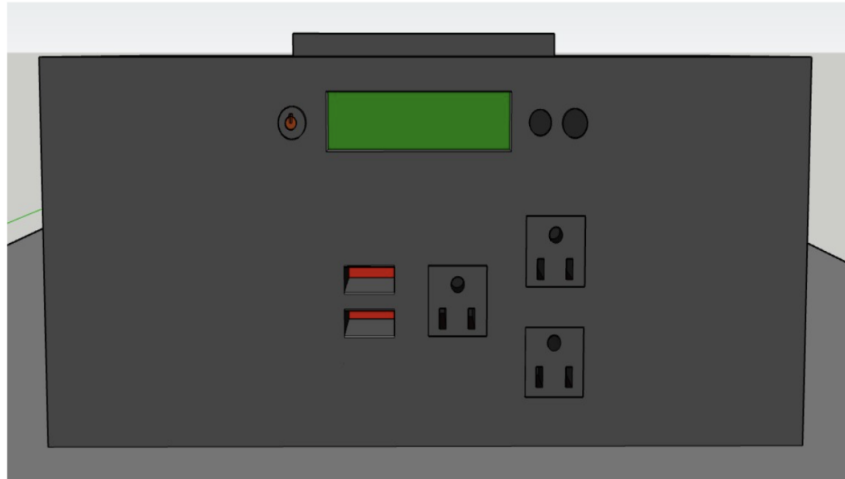


Hardware Design

The 3D representation shows the overall design of the detachable power source and the solar-powered rental station. It provides a clear visual of the system's shape and structure. The model illustrates how the different parts of the system are arranged in space, allowing a better understanding of the physical layout of the proposed system.

Figure 3.9

Front View of the Power Source – Three Dimensional Representation

**Figure 3.10**

Aerial View of the Power Source – Three Dimensional Representation



Figure 3.11

Three Dimensional Representation of the Rental Station of the Detachable Power Source



Figure 3.15 shows the 3D design of the system. It is a stationary power station powered by solar energy and contains three detachable power sources. The system has a single coin slot that controls access, and each detachable power source is secured with a solenoid lock.

3.1.4 Software Development

This part provides an outline of the software development for the solar-powered rental station with detachable power sources. The primary objective of the software is to manage the rental, monitoring, and control of portable power units through data acquisition, processing, and decision-making. The system integrates embedded system programming, mobile application development, and cloud-based data management to ensure reliable, automated, and secure operation.

The software is designed to monitor battery status, track rental duration, determine location coordinates, and calculate user credits based on predefined system rules. These functionalities allow the system to support automated unlocking and locking mechanisms, notification handling, and synchronization between hardware, mobile application, and cloud services.

Embedded System Software Development

This section discusses the development of the embedded software responsible for data acquisition, processing, and hardware control. An ESP32 microcontroller serves as the main processing unit due to its built-in wireless communication capabilities, low power consumption, and suitability for Internet of Things (IoT) applications.

The embedded software continuously collects data from the battery monitoring circuit, GPS module, and GSM communication module. The acquired data are processed locally to determine battery status, rental duration, and location of the detachable power source. Processed information is then transmitted wirelessly to the cloud server and mobile application for monitoring and record-keeping.

Battery Monitoring and Power Calculation

Battery voltage is measured using the ESP32's analog-to-digital converter (ADC). The measured voltage values are mapped to corresponding battery percentage levels using predefined threshold ranges. Based on the battery percentage at the start and end of the rental period, the system calculates the amount of unused battery power.

The unused power is used as the basis for computing user credits, encouraging responsible usage and timely return of the power units.

Rental Time Tracking and Rule-Based Control

Rental duration is tracked using internal system timers. The software enforces a maximum rental duration of twelve (12) hours per transaction. Time-based conditions are implemented to trigger system actions, such as sending notification alerts before the rental deadline and recording violations

after the rental period expires.

Rule-based logic controls critical events, including unlocking the detachable power source after successful payment verification and locking the unit upon return confirmation.

Mobile Application Development

This section presents the development of the mobile application that serves as the primary user interface of the system. The mobile application enables users to monitor battery status, view location, track rental duration, receive notifications, and check earned credits.

The application communicates with the embedded system and cloud database through wireless connectivity, ensuring synchronization and accurate data presentation.

Flutter Framework

Flutter was selected as the development framework for the mobile application due to its cross-platform capability, high performance, and rich user interface components. Flutter employs the Dart programming language, which is compiled directly into native machine code, allowing fast execution and smooth user interactions.

Using Flutter enabled the developers to create a single codebase that

can be deployed across multiple platforms. The framework's extensive widget library facilitated the development of an intuitive and responsive user interface for displaying system data such as battery level, rental status, notifications, and user credits. The hot reload feature also supported rapid development, testing, and refinement of the mobile application.

Activity Diagram

Figure 3.12

Activity Diagram of the Community Resident

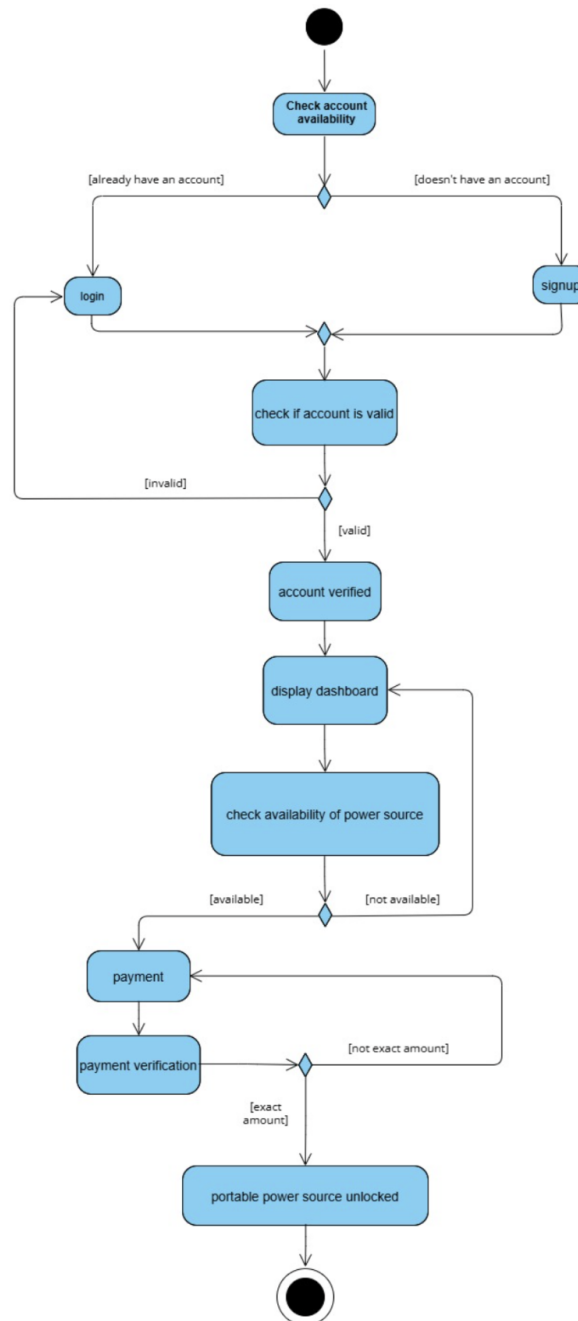


Figure 3.12 shows the step-by-step activity flow of the system from user login to unlocking the portable power source. The process begins with checking account availability. If the user already has an account, they proceed to the login step; otherwise, they must sign up first. After logging in, the system checks if the account is valid. If the account is invalid, the process returns to the login step. If valid, the system verifies the account and displays the dashboard. Next, the system checks if a power source slot is available. If no slot is available, the user remains on the dashboard until one becomes free. If a slot is available, the process continues to the payment step. The system then verifies the payment amount. If the amount entered is not exact, the process returns to the payment step. If the payment is correct, the portable power source is successfully unlocked, completing the process.

Overall, the diagram clearly shows how the user interacts with the system, which starts from account access, validation, and payment, up to unlocking the rented portable power source.

GitHub

GitHub was utilized for version control and collaborative development of the mobile application. All source code, documentation, and configuration files were maintained in a centralized GitHub repository, allowing multiple de-

developers to work concurrently while tracking changes efficiently. GitHub enabled version tracking, where all revisions to the code were logged and could be reverted if necessary. The use of branching and merging allowed new features and fixes to be developed in isolated branches and later integrated into the main branch following review. Additionally, GitHub facilitated team collaboration by providing mechanisms for code review, pull requests, and task management through GitHub issues. The repository also ensured a secure and accessible backup of the project, which could be accessed from any location with an internet connection. The integration of GitHub into the development workflow improved code quality, reduced conflicts, and supported smooth project management throughout the software development process.

Cloud Database and Backend Services

Cloud-based services are utilized to support data storage, transmission, and system synchronization. These services enable secure handling of rental transactions, user records, and system logs.

Firebase Firestore

Firebase Firestore is used as the primary cloud database for storing structured system data. This includes user profiles, rental transactions, battery usage records, rental timestamps, and earned credits. Firestore's data

synchronization capability allows instant updates across the embedded system and mobile application, ensuring data consistency and reliability.

System Measurements and Calculations

This section presents the specific measurements and calculations performed by the software to support rental management, power monitoring, and credit computation. These calculations are derived from sensor readings, system timestamps, and predefined rules integrated into the software logic.

Battery Voltage Measurement

The ESP32 continuously measures the battery voltage using its analog-to-digital converter (ADC). The measured ADC value is converted into the actual battery voltage using a voltage scaling factor based on the battery monitoring circuit.

This voltage value serves as the basis for determining battery percentage, power availability, and unused power.

Battery Percentage Calculation

Battery percentage is calculated by mapping the measured battery voltage to predetermined minimum and maximum voltage limits of the LiFePO_4 battery.

Rental Time Computation

Unused Power Calculation

Credit Computation

Location Measurement and Tracking

The GPS module provides latitude and longitude values of the detachable power source. These coordinates are transmitted to the system and displayed in the mobile application to indicate the unit's current location.

This location data supports monitoring, recovery, and security operations.

Slot Availability and Lock Status Determination

Slot availability is determined based on the presence of the detachable power source and rental state:

- Free

Unit is available, Locked

- Occupied

Unit is currently rented, Unlocked

The lock status is controlled through software logic and depends on payment validation, and return confirmation.

3.1.5 System Integration

3.1.6 Testing

3.1.7 Implementation

The research will be conducted in selected areas of Camiguin and Misamis Oriental, regions in the Philippines that are known for frequent and unscheduled power interruptions. These areas are served by CAMELCO (Camiguin Electric Cooperative) and MORESCO II (Misamis Oriental Electric Cooperative II), both of which experience regular power outages, particularly during peak hours. This issue disrupts daily activities, especially in community centers, educational institutions, and local businesses that rely on electricity for mobile devices, lighting, and small appliances.

3.1.8 Data Collection

The research team will gather data from both primary and secondary sources. For the primary data, first-hand interactions will be conducted with community residents, power supply providers, and solar energy experts. Community residents, as the primary users of the solar-powered rental station, will provide information on the demand for portable power sources, willingness to adopt a solar-powered system, and feedback on system usability and effectiveness in addressing power interruptions. Power supply providers, such as local

electric cooperatives like CAMELCO and MORESCO 2, will provide data on the frequency of power interruptions, challenges in accessing reliable energy, and the overall energy demand in the community. Insights from solar energy experts will focus on technical aspects of solar power systems, including optimal photovoltaic panel configurations, energy storage solutions, and power inverter specifications.

Secondary sources will include academic papers, case studies, industry reports, and other literature related to solar energy, portable power stations, and renewable energy systems. These sources will provide a theoretical foundation and context for the study, ensuring that both practical relevance and technical feasibility are considered in developing the solar-powered rental station system.

3.1.9 Data Gathering Procedure

To obtain relevant qualitative data for the study, a series of in-depth interviews will be conducted with selected community residents using purposive sampling. The interviews will focus on understanding residents' experiences with frequent power interruptions, the challenges they encounter during interruptions, and their perceptions of a solar-powered rental station with detachable power source as an alternative energy source. These discussions will also

explore their reliance on backup power systems and their willingness to adopt renewable energy solutions. Participants will be purposively selected, particularly those who frequently experience power interruptions, depend on electricity for livelihood activities, and have expressed interest in solar technology. Combined with qualitative tools, this approach will ensure a comprehensive understanding of the community's energy needs, user expectations, and potential areas for improving the design and implementation of the solar-powered rental station system.

3.1.10 Data Finding Analysis

The analysis of the collected data will lead to several important findings. The first will reveal that community residents frequently experience power interruptions that disrupt both household and livelihood activities. These interruptions will highlight common challenges related to the reliability, cost, and accessibility of backup power solutions. Residents will express concerns about their dependence on the unstable power grid and the lack of affordable alternatives during interruptions. The analysis will also uncover the community's openness to adopting a solar-powered rental solution, recognizing its potential to provide a more sustainable and reliable energy source. Feedback from interviews and discussions will emphasize the need for practical improvements

to the proposed system, such as flexible payment options (coin- or app-based), increased power capacity, and user-friendly accessibility. These insights will be vital in assessing the system’s feasibility and refining its design to ensure it effectively addresses the residents’ energy needs.

3.1.11 User Definition

After the planning phase, the system’s users were identified. For this study, the Community Resident is defined as follows:

Community Resident - A person residing in a local community who is actively involved or has access to the portable power sources during power interruptions or emergencies.

3.1.12 System Requirements

Table 3: System Requirements

Category	System Requirement
Input Requirements	- The system shall collect user information during account registration through designated input fields in the mobile application. - The system shall accept user login credentials such as username and password for authentication. - The system shall accept rental requests, including power source selection, rental duration, and payment method (coin-based or app-based). - The system shall receive telemetry data from each power source, including battery percentage, GPS location, and rental status through the GSM module.
Continued on next page	

Table 3 – continued from previous page

Category	System Requirement
	- The system shall receive telemetry data from each power source, including battery percentage, GPS location, and rental status through the GSM module.
Process Requirements	- The system shall process rental transactions by verifying payment completion before authorizing power source access. - The system shall generate and transmit an access token or unlock code to the power source upon successful rental approval. - The system shall continuously monitor and record the battery level, location, and usage duration of active rentals. - The system shall automatically send an SMS notification via GSM one hour before the allotted 12-hour usage period ends, and if the user exceeds the 12-hour limit, it shall continue sending overdue alerts every 10 minutes until the item is returned. - The system shall process the return of rented power sources by validating device ID, updating the database, and releasing the final transaction summary. - The system shall calculate credit earned for users based on unused power and reflect the updated balance in the users mobile application account.
Output Requirements	- The system shall display rental confirmation details such as power source ID, rental duration, and current charge level. - The system shall provide updates on the detachable power source, provide user logs, and notify the admin whenever a new account is created. - The system shall generate the location of the rented power source using the GSM module. - The system shall generate notifications on the mobile app, including GPS location, and rental time remaining.
Continued on next page	

Table 3 – continued from previous page

Category	System Requirement
	- The system shall calculate the unused power of the returned detachable power source and convert it into a credit stored in the user's mobile app.
Control Requirements	- The system shall implement user authentication and authorization mechanisms to restrict access to registered users only. - The system shall validate all rental and payment data to ensure that only completed and legitimate transactions are processed.
Performance Requirements	- The system shall maintain fast response time during login, payment processing, and rental activation to ensure a smooth user experience. - The system shall provide monitoring and synchronization of power source data through reliable GSM and IoT communication. - The system shall ensure continuous operation and high availability to prevent service interruptions during rentals.

3.1.13 Calculation Requirements

Load Analysis

Table 4: Load Consumption

Load	Quantity	Power	Hours/Day	Daily Consumption
Electric Fan	1	75 W	5	370 WH
Mobile Phone Charger	1	20 W	5	100 WH
Auxiliary Load	1	5 W	5	25 WH
Total				500 WH

The schedule of load includes three main loads, namely the electric fan,

the mobile phone charger, and a 5W auxiliary load. Each load is multiplied by its hours of operation, and their products are summed to obtain the total watt-hour daily consumption.

All calculations are done in accordance to PEC (Philippine Electrical Code 2017).

Battery Sizing

- **Detachable Power Source:**

$$\text{Battery Capacity} = \left(\frac{\text{Total Daily Consumption}}{\text{System Voltage}} \right) + \text{ESP32 Consumption} \quad (1)$$

This equation determines the required battery capacity for the detachable power source. The total daily load consumption in watt-hours is divided by the 12V system voltage to convert the value into ampere-hours. The consumption of the ESP32 microcontroller (0.858 Ah), which continuously operates for system monitoring, is then added. This computation ensures that the detachable power source can sustain both the primary load and the microcontroller's operational requirements. Please refer to Appendix A for the detailed computation.

- **Station:**

$$\text{Battery Capacity} = (\text{BCDPS} + 35\%) + \text{ESP32 Consumption} \quad (2)$$

This equation is used in sizing the battery of the main station. It begins with the computed battery capacity of the detachable power source and adds 35% to account for system losses, such as wiring resistance, power conversion inefficiencies, and charging losses. The ESP32's consumption (0.858 Ah) is also included to ensure that the main station battery can support all monitoring and control operations. Please refer to Appendix A for the detailed computation.

- **Panel Sizing:**

$$\text{PV Power} = \frac{\text{Total Daily Consumption}}{\text{Sun Peak Hours}} \quad (3)$$

This equation calculates the required power rating of the solar panel. The total daily energy consumption (500 Wh) is divided by the available sun peak hours (3.5 hours, based on PEC 2017) to determine the power output needed to fully recharge the battery each day. The computed value of approximately 142.86W is rounded up to a 200W solar panel for practical and technical considerations. Please refer to Appendix A for the detailed computation.

- **Inverter Sizing:**

Table 5: Inverter Sizing

Load	Power	Power Surge	With Power Surge
Electric Fan	75 W	3	225 W
Mobile Phone Charger	20 W	1	20 W
Auxiliary Load	5 W	1	5 W
Total			250 W

$$\text{Inverter Size} = \text{Total Power Consumption} + 20\% \text{ Allowance} \quad (4)$$

This formula is used to determine the appropriate inverter size. After calculating the total power requirements of all connected loads, including surge components such as the electric fan, a 20% allowance is added to ensure that the inverter can safely handle fluctuations and temporary increases in power demand. With a total of 250W, the system requires a 300W pure sine wave inverter. Please refer to Appendix A for the detailed computation.

- **Solar Charge Controller Sizing:**

$$I_{cc} = \frac{\text{Total PV Power}}{\text{System Voltage}} \quad (5)$$

This equation determines the required rating for the solar charge controller. Dividing the total solar panel wattage (200W) by the system

voltage (12V) yields the expected charging current supplied to the controller. The computed value of 16.67A is rounded upward to select a 20A MPPT charge controller, ensuring safe and efficient power regulation. Please refer to Appendix A for the detailed computation.

- **Breaker Sizing:**

- a. Panel - SCC

$$\text{Circuit Breaker} = 11.85A \times \text{Over Radiance Factor} \times \text{Safety Factor} \quad (6)$$

This equation is used to determine the appropriate circuit breaker rating between the solar panel array and the Solar Charge Controller (SCC). The value 11.85A represents the short-circuit current (Isc) of the solar panel system. Multiplying by the Over Radiance Factor and the Safety Factor provides an adjusted current value that accounts for sudden increases in sunlight intensity and unexpected current surges. After substituting the given parameters in the complete computation, the calculated value results in a 20A circuit breaker rating, which is the recommended protection size between the solar panel and the SCC. This ensures safe operation by preventing overheating, protecting the solar wiring, and avoiding

damage to system components. Please refer to Appendix A for the detailed computation.

b. SCC - Battery

$$\text{Circuit Breaker} = \frac{\text{Total Array Wattage}}{\text{Battery Bank Nominal Voltage}} \times \text{Safety Factor} \quad (7)$$

This equation sizes the circuit breaker between the Solar Charge Controller (SCC) and the battery bank. The total wattage of the solar panel array is divided by the system's nominal battery voltage to determine the expected charging current that will flow from the SCC into the battery during operation. A Safety Factor is then applied to account for possible current variations during charging, ensuring that the system remains protected from overload. After substituting the system values into the calculation, the resulting breaker rating requirement is 20A, which provides proper protection during charging while maintaining safe operating conditions for the battery bank. This recommended 20A circuit breaker prevents overheating and electrical damage by limiting excessive current flow into the battery storage system. Please refer to Appendix A for the detailed computation.

c. Battery - Power Inverter

$$\text{Circuit Breaker} = \frac{\text{Power Inverter}}{\text{Battery Bank Nominal Voltage}} \quad (8)$$

This equation determines the appropriate circuit breaker rating between the battery bank and the power inverter. The power rating of the inverter (in watts) is divided by the nominal battery voltage (in volts) to calculate the maximum current drawn during inverter operation when supplying AC loads. This calculated current serves as the basis for selecting a circuit breaker capable of protecting both the inverter and battery system during high-demand usage. After substituting the specified system values into the equation, the resulting breaker requirement is 80A, which is the recommended rating to ensure sufficient protection from overcurrent conditions that may occur when connected appliances operate at maximum load capacity. This breaker rating prevents damage to the inverter and battery by interrupting excessive current flow and maintaining safe operating conditions. Please refer to Appendix A for the detailed computation.

d. Inverter - Load (AC Breaker)

$$\text{Circuit Breaker} = \frac{\text{Safety Load}}{\text{Inverter Voltage Output}} \times \text{BBNV} \quad (9)$$

This equation sizes the AC circuit breaker located between the inverter and the connected load. The Safety Load represents the maximum expected power consumption of the AC appliances connected to the inverter. Dividing this value by the inverter's AC output voltage converts the load into current, representing the expected amperage during operation. The result is then multiplied by the battery bank nominal voltage to account for conversion losses and variations in real operating conditions. Based on the system values used in the computation, the required AC breaker rating is 4A, which provides sufficient protection by interrupting excessive current that could damage connected appliances or the inverter output circuitry. This calculated breaker size ensures safe and reliable system performance under normal and peak load operations. Please refer to Appendix A for the detailed computation.

- **Wire Sizing:**

$$\text{Voltage Drop Index} = \frac{\text{Amperes} \times \text{Wire Length (Feet)}}{\text{Voltage} \times \text{Percent Voltage Drop}} \quad (10)$$

This equation is used for determining the appropriate wire size needed to safely and efficiently handle electrical current within the system. The Voltage Drop Index (VDI) is calculated based on four factors: the cur-

rent in amperes, the wire length, the system voltage, and the allowable percentage of voltage drop. Higher current flow or longer cable distance increases the VDI value, which indicates the need for a thicker wire in order to minimize electrical resistance and heat generation. Selecting the correct wire size ensures that voltage loss is controlled, system performance remains efficient, and the overall electrical design operates safely without risk of overheating or component failure.

a. Panel - SCC

Panel to Solar Charge Controller = 10AWG

The first VDI calculation was performed for the wire connection between the solar panel and the Solar Charge Controller (SCC). Given the operational current and wire distance in this segment, the computed VDI value resulted in 3.52. Based on standard VDI wire reference charts, a value of 3.52 corresponds to the use of 10 AWG wire, which provides sufficient conductivity and prevents excessive voltage reduction over the cable length. This ensures that power harvested from the solar panel is efficiently delivered to the SCC with minimal losses. Please refer to Appendix A for the detailed computation.

b. SCC - Battery

Solar Charge Controller to Battery = 10AWG

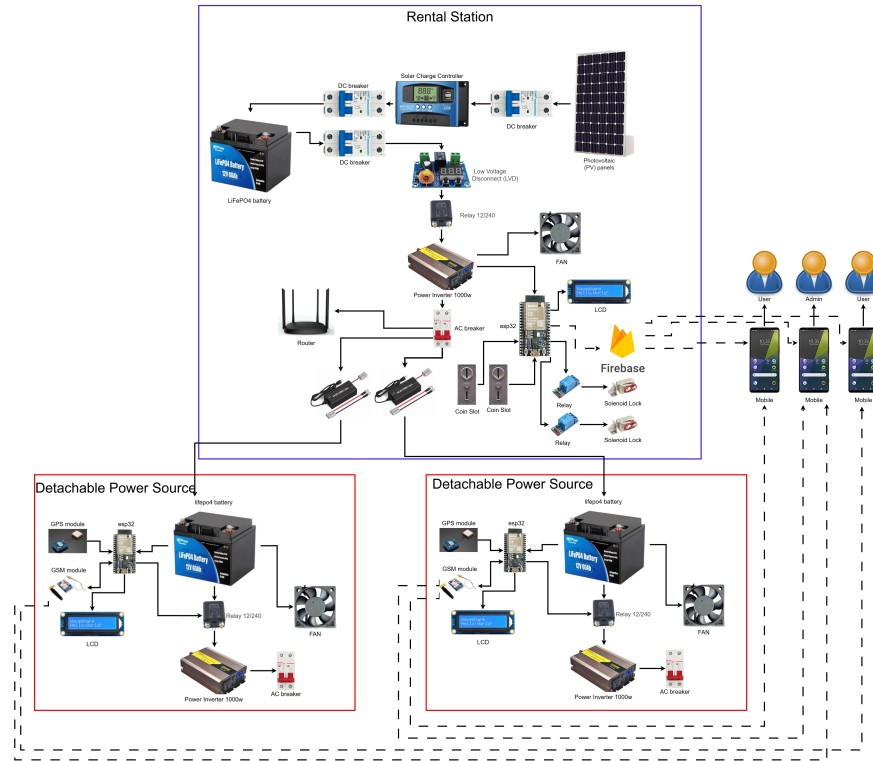
The second VDI calculation was conducted for the wire connection between the Solar Charge Controller (SCC) and the battery bank. Using the specified current, distance, system voltage, and allowable voltage drop percentage, the computed VDI value for this segment is 3.125. Similar to the previous result, this VDI value also aligns with the requirement for 10 AWG wire, ensuring safe current handling capacity and controlled voltage drop. The consistent selection of 10 AWG wire across both connection points maintains electrical balance and reliability throughout the charging circuit. Please refer to Appendix A for the detailed computation.

3.2 Technical Design Workflow

3.2.1 System Architecture

Figure 3.13

System Architecture



The Figure 3.3 illustrates the overall architecture of the solar powered rental station with a detachable power source. The system is designed to harvest solar energy, convert it to usable power, and provide access to users through various interfaces and technologies.

- *Solar Energy Collection and Storage* - PV panels capture solar energy,

which is stored in a LiFePO_4 battery through a solar charge controller and DC breakers for protection.

- *Energy Conversion* - The stored DC power is converted to AC using a power inverter to supply household appliances.
- *User Interface and Control* - The ESP32 microcontroller connects with a mobile app via a router and Firebase, allowing users to monitor battery levels and manage rental requests.
- *Power Access and Management* - A coin slot and relay control access to the power source, while an LCD displays system status.
- *GPS Tracking and Security* - A GPS and GSM module track the power source's location, ensuring security and preventing theft.
- *Admin Control* - Admins can monitor and manage the system through a mobile interface, overseeing usage and system health.

This system is designed to provide a sustainable, secure, and user-friendly power source solution for communities, utilizing solar energy and modern IoT technologies for enhanced management and accessibility.

3.2.2 Flowchart

Figure 3.14

System Flowchart

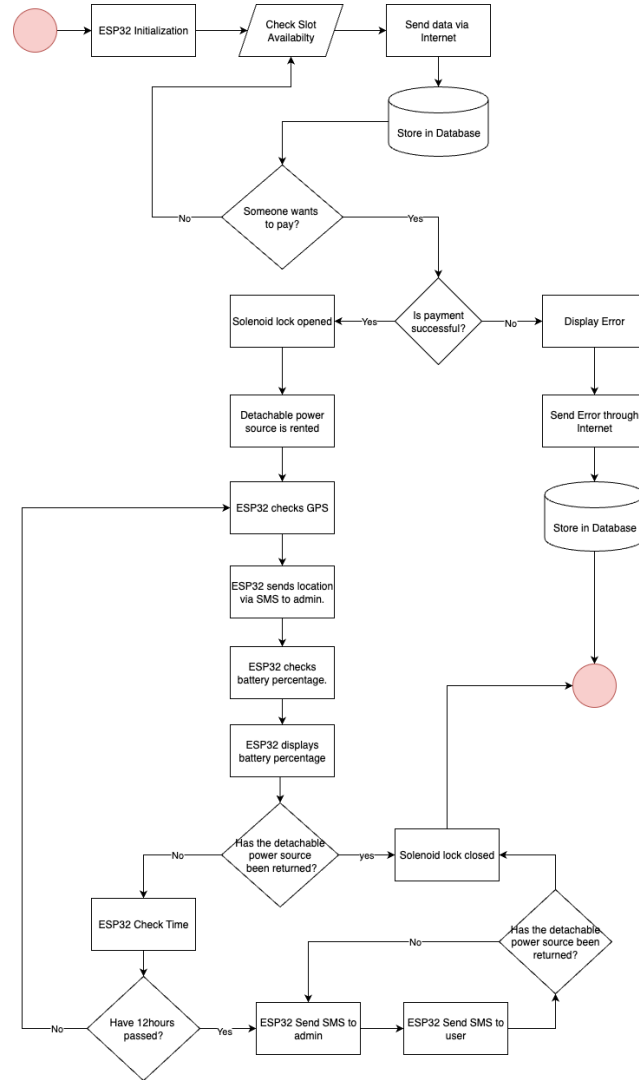


Figure 3.4 illustrates the overall process of the proposed Solar-Powered Rental Station with Detachable Power Sources. The process begins with the initialization of the ESP32, which activates system components and prepares

communication protocols. The system first checks slot availability to determine if a detachable power source is ready for use. Data from this process is transmitted through the internet and stored in a central database for monitoring purposes.

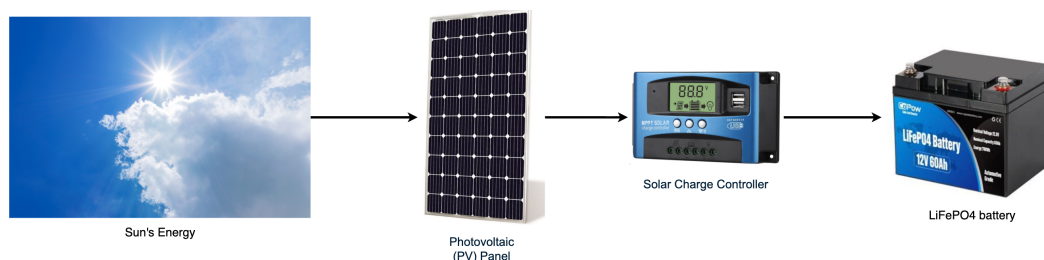
When a user intends to rent a power source, the system verifies if payment is successful. If confirmed, the solenoid lock opens, allowing the detachable power source to be accessed by the user. Once rented, the ESP32 continuously monitors the power source, including its GPS location and battery percentage, and sends the information to the administrator via SMS. This ensures proper tracking and security of the rented device.

The system also monitors whether the detachable power source is returned. If it remains unreturned after 12 hours, the ESP32 automatically sends reminder SMS notifications to both the user and the administrator. Upon return, the solenoid lock closes, securing the power source back in place. All key actions, including errors and system activities, are logged in the database for reference and analysis.

3.2.3 Solar Energy Harvesting

Figure 3.15

Conversion of Sunlight into Electrical Energy using PV Panels and a Charge Controller



The Figure 3.5 illustrates the process of harvesting solar energy using photovoltaic (PV) panels. Solar energy is captured by the PV panels, which then convert sunlight into electrical energy. The energy is stored in a LiFePO_4 battery through a solar charge controller, ensuring proper charging and voltage regulation. This energy is stored for later use, contributing to the overall sustainability and efficiency of the system, enabling it to power devices even during periods of limited sunlight.

3.2.4 Energy Conversion DC to AC

Figure 3.16

Battery DC power converted to AC output via Inverter.

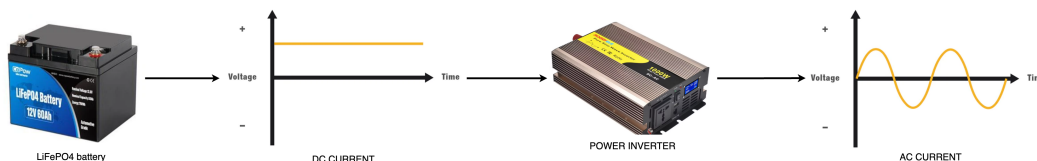
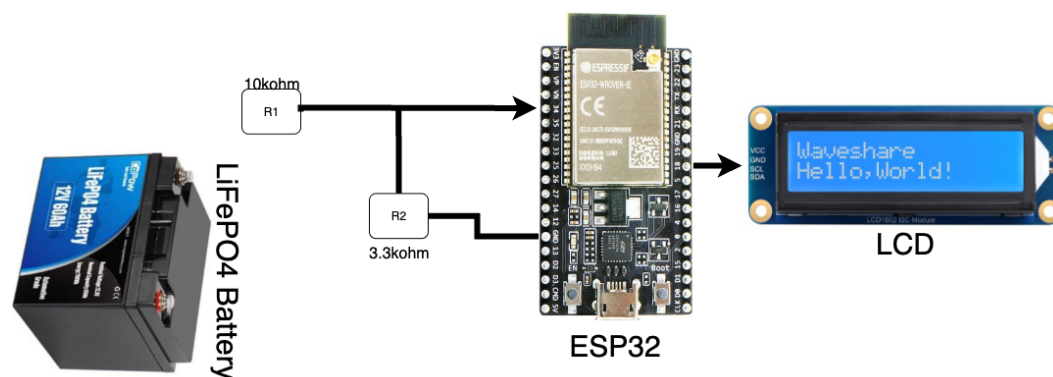


Figure 3.6 shows the process of converting direct current (DC) into alternating current (AC). The stored energy in the LiFePO_4 battery is passed through a power inverter, and the inverter transforms the DC from the battery into AC power, which can be used to power household appliances such as fans and other AC-powered devices. This conversion is essential for making the system compatible with common electrical appliances that require AC power for operation.

3.2.5 Power Monitoring

Figure 3.17

ESP32 measures battery voltage and displays data on LCD



The Figure 3.7 illustrates the power monitoring system used to keep track of the energy stored and consumed by the system. The LiFePO_4 battery is connected to an ESP32 microcontroller, which continuously monitors the battery's voltage and health, and the data collected is displayed on an LCD

screen which allows users to view the battery's current status. This monitoring is vital for ensuring the system's efficiency and preventing over-discharge, ensuring long-term sustainability.

3.2.6 GPS Tracking

Figure 3.18

ESP32 receives location data from GPS satellites using GSM Module

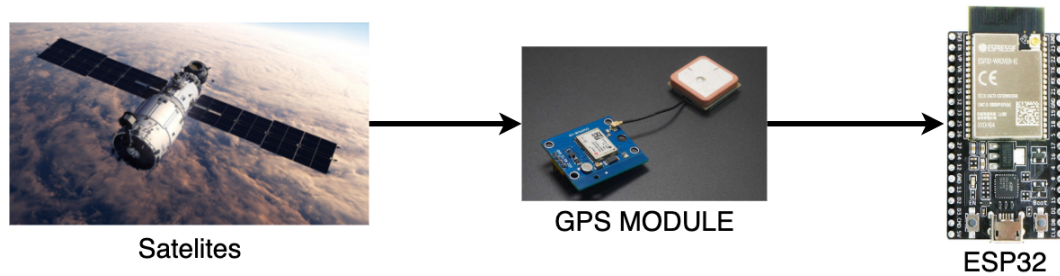
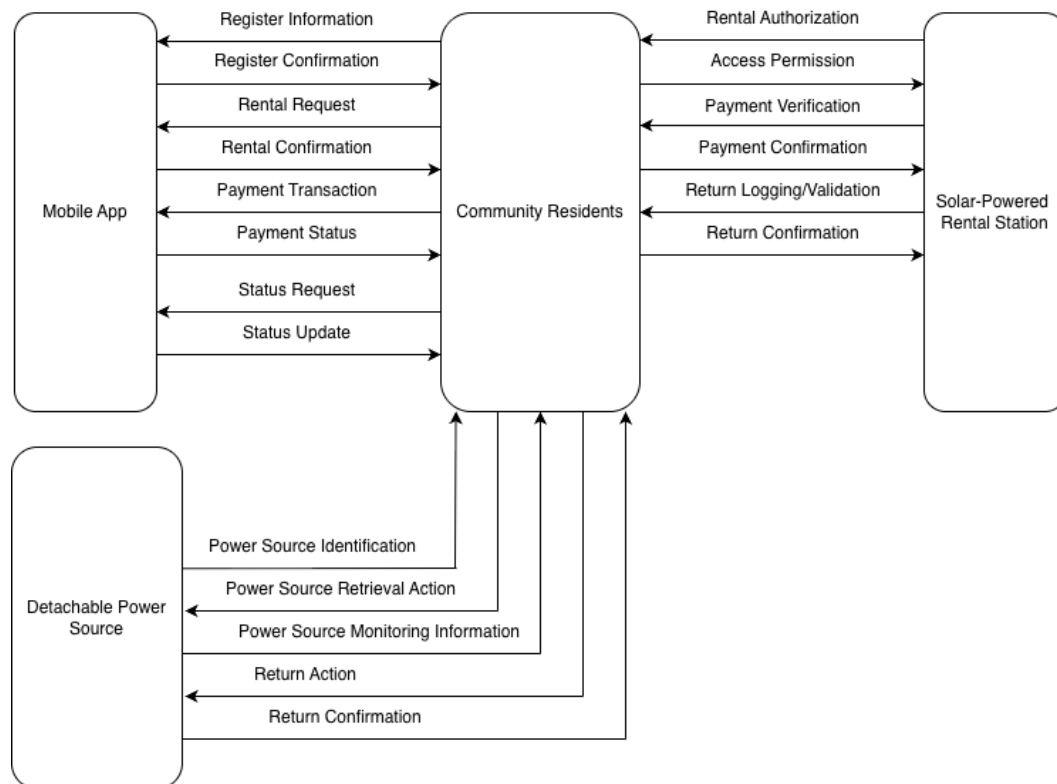


Figure 3.8 shows the GPS tracking system used to monitor the location of the detachable power source. A GPS module, connected to the ESP32 microcontroller, receives signals from satellites to determine the geographical location of the system. This location data is essential for tracking and managing the rental and use of the power sources, ensuring they are accessible to users and protected from theft or misuse. The GPS data is then sent to the system for further processing and user access.

3.2.7 Context Level Diagram

Figure 3.19

Context Level Diagram



This Figure 3.9 illustrates the complete interaction flow of the solar-powered rental system, showing how the Mobile App, Community Residents, Solar-Powered Rental Station, and the Detachable Power Source communicate with one another. The process begins with community residents using the mobile application to register their information, submit rental requests, process payments, and receive status updates. Once a rental request is made, the

app sends the relevant information to the system, which then confirms the registration, validates the rental request, and updates the user about payment status and rental confirmation.

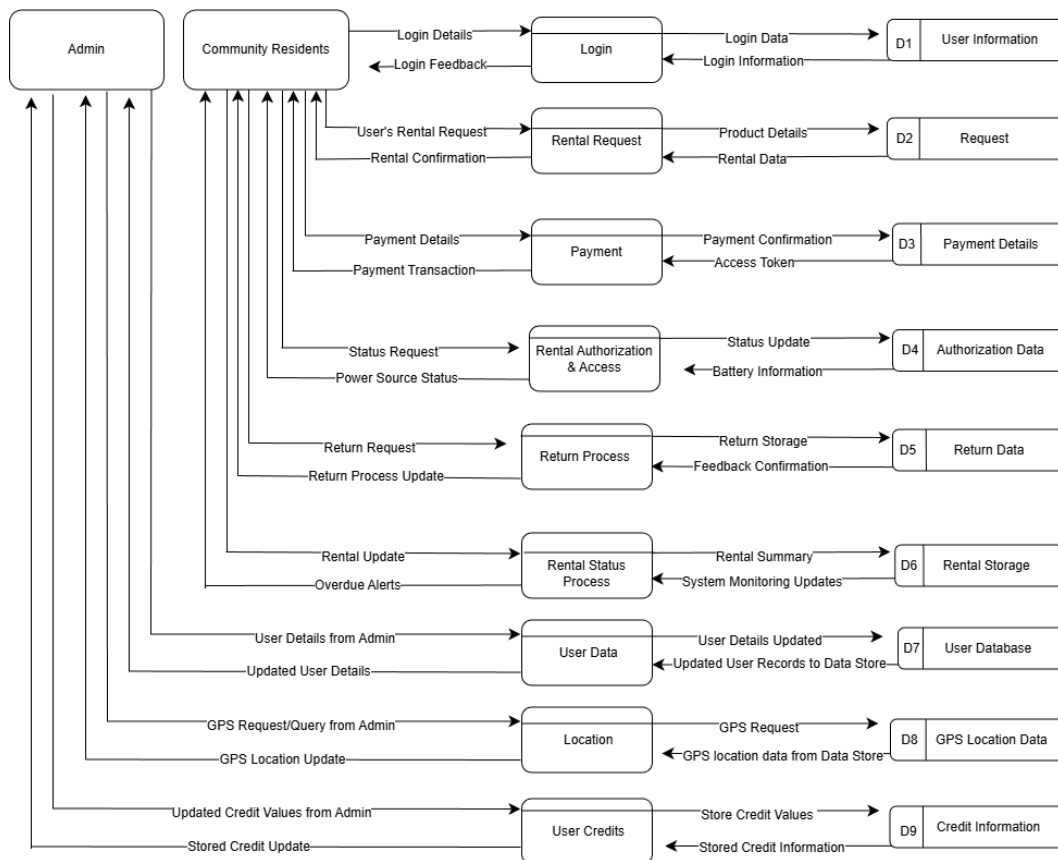
The interaction between the community residents and the solar-powered rental station focuses on physical access and transaction validation. The rental station authorizes the rental, verifies payment, and grants access permission for users to retrieve the detachable power source. Upon returning the device, the station logs and validates the return, ensuring that the power source is securely and correctly returned. The station then issues a return confirmation to the user, completing the rental cycle.

The detachable power source also communicates with the system to strengthen security and monitoring. It provides identification data, sends monitoring information such as location and usage status, and confirms both retrieval and return actions. These interactions ensure that the system can track each device, prevent unauthorized usage, and maintain accountability for each rental unit. Overall, the diagram presents a structured overview of how all components work together to support registration, rental, payment, monitoring, and return of the solar-powered power source in a seamless and secure manner.

3.2.8 Data Flow Diagram

Figure 3.20

Data Flow Representation of the Solar-Powered Rental Station System



The Figure 3.10 shows the complete process flow of the solar-powered rental station system, illustrating how community residents interact with the mobile application and system database. It begins with the login process, where residents input their login details, which are verified by the system. Once authenticated, the system provides feedback confirming successful login.

After logging in, the user proceeds to make a rental request by selecting a power source and specifying the rental duration. The system responds with rental confirmation and product details. The user then moves to the payment process, where payment details are submitted through the app. The system validates the payment, sends a confirmation, and generates an access token that authorizes the user to unlock and use the detachable power source.

Once access is granted, the rental authorization and monitoring phase begins. The system continuously provides updates on the power source's status, such as battery level and remaining usage time, ensuring that the user can track energy consumption. When the rental period ends, the user then sends a return request, and the system verifies and confirms the return while updating the rental status. After returning the power source, the user proceeds to the feedback process, submitting performance feedback through the app. The system stores the feedback and confirms successful submission. After that, the system records the final rental summary and sends any overdue alerts if the power source is not returned on time. Overall, the diagram presents a clear and structured view of how the user's actions and system responses are linked throughout the entire rental transaction cycle, which is from login to feedback completion.

3.2.9 Use Case Diagram

Figure 3.21

Use Case Diagram

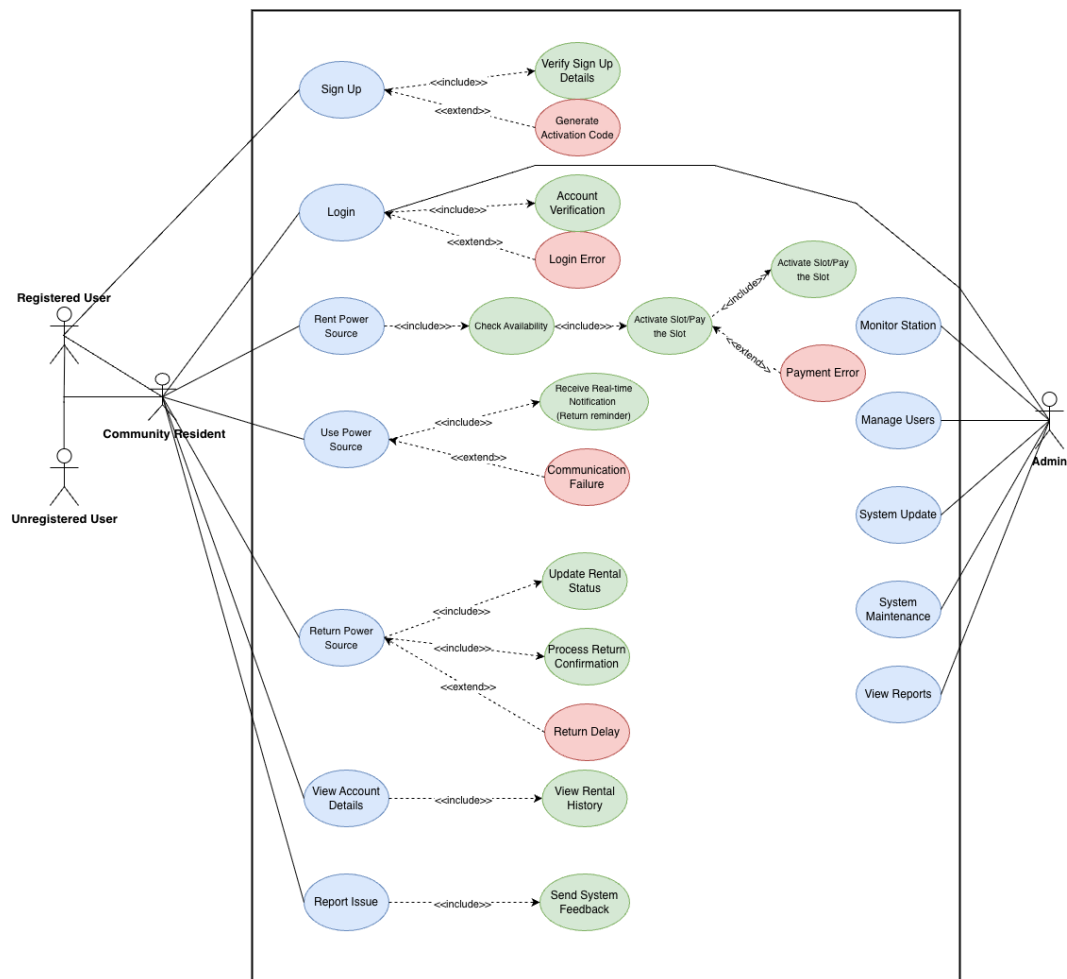


Figure 3.11 illustrates how different users and administrators interact with the system. It highlights the actions users can take and the roles of administrators. The Unregistered User is someone who has not yet signed up for the system. They can create an account by providing their details, which

are then verified by the system. If the details are correct, an activation code is generated for the user to complete the registration. After signing up, the user can log in to the system, but if there's an issue with the login, an error is displayed.

Once the Unregistered User becomes a Registered User after logging in successfully, they can perform more actions within the system. The user can rent a power source by checking its availability, paying for it, and activating the slot where the power source is stored. If there's a problem with communication during the rental process, the system handles this as an extended action. After using the power source, the user can return it, and the system confirms the return. If there's any delay in returning the power source, the system will manage this as well. Registered users also have the ability to view their account details and rental history, and they can report any issues or provide feedback through the system. The Admin plays a key role in overseeing the system. They can manage users, including adding or removing them from the system. The admin is also responsible for performing system updates and maintenance, ensuring the smooth operation of the system. Additionally, the admin can view various reports that provide insights into system performance and user activities.

The relationships in the diagram are shown through Include and Ex-

tend. Include means that one action is always linked to another. For example, signing up always includes verifying the user's details, and renting a power source includes checking availability and processing payment. Extend means that certain actions may trigger additional steps under specific conditions. For instance, if a payment fails, it will trigger the "Payment Error" action, or if the power source return is delayed, it will trigger the "Return Delay" action. In summary, this use case diagram provides an overview of how users and administrators interact with the system, covering key actions like renting and returning power sources, managing user accounts, and maintaining the system.

3.2.10 Mobile App

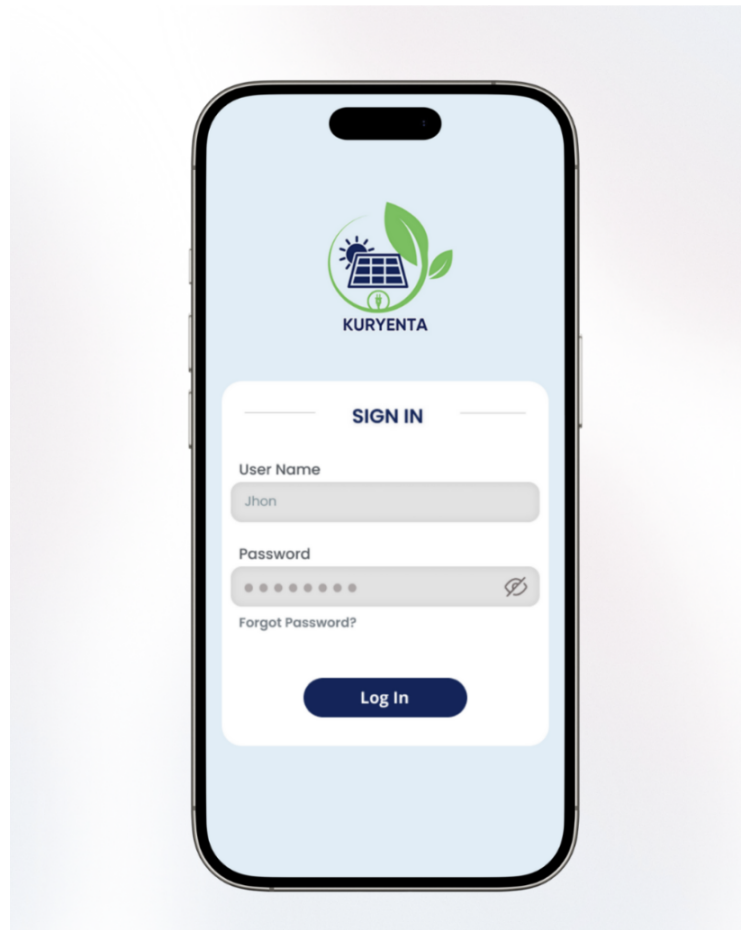
Figure 3.22

Welcome Page



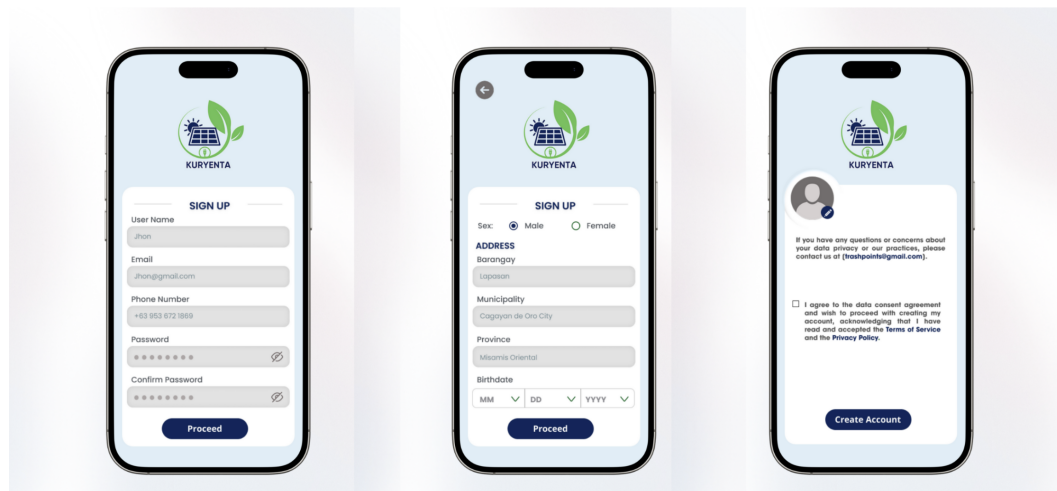
The welcome interface presents the system logo and entry actions. The left panel shows a splash screen with a “Get Started” control; the right panel shows a gateway card offering Log In and Sign Up. This screen functions as the access point to the authentication flow.

Figure 3.23
Sign in Screen

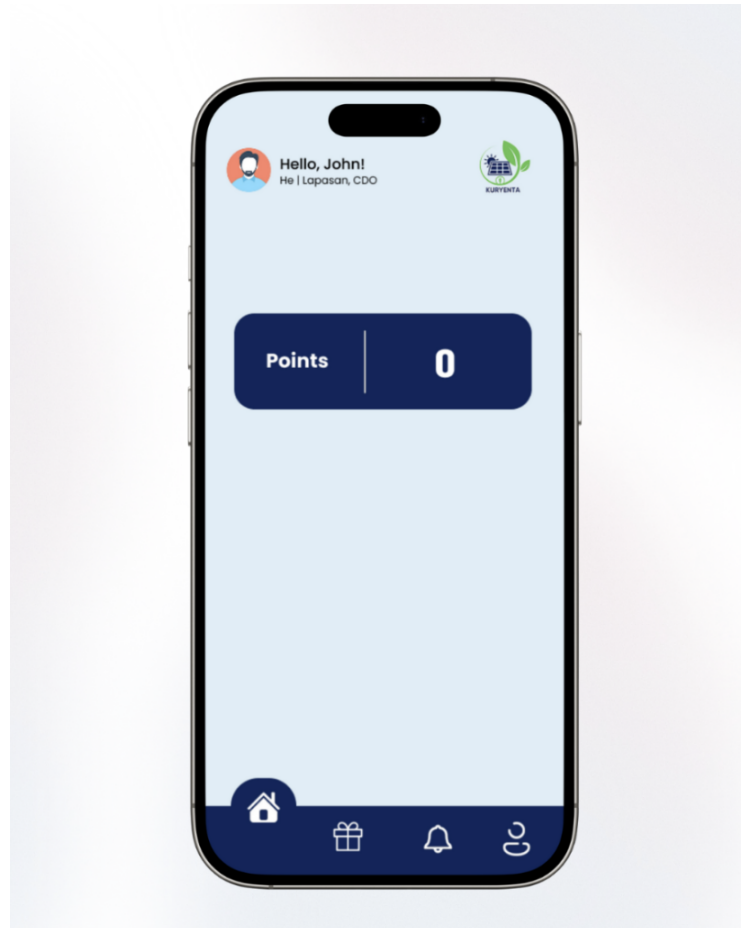


The sign-in interface collects the username and password with options to reveal the password and recover forgotten credentials. A primary Log In control initiates authentication for registered users. The design supports secure access to user functions.

Figure 3.24
Sign Up Screen

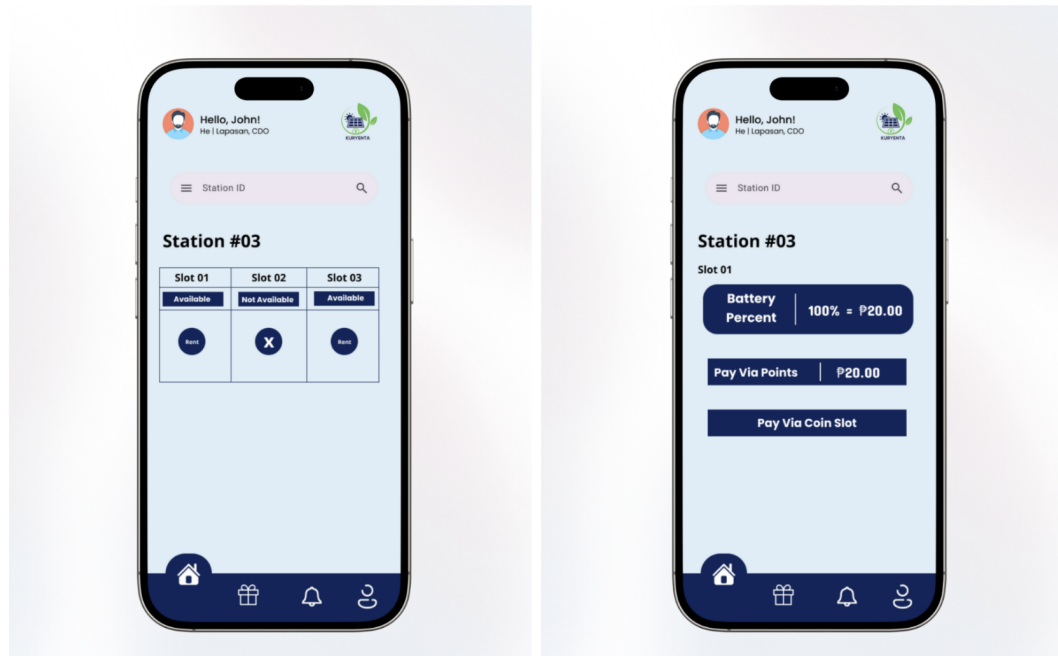


Registration is implemented as a three-step process. Step 1 captures account credentials (username, email, phone number, password, and confirmation). Step 2 records profile and address data (sex, barangay, municipality, province, and birthdate). Step 3 presents a data-privacy notice and obtains consent to the Terms of Service and Privacy Policy before account creation.

Figure 3.25*User Home Screen*

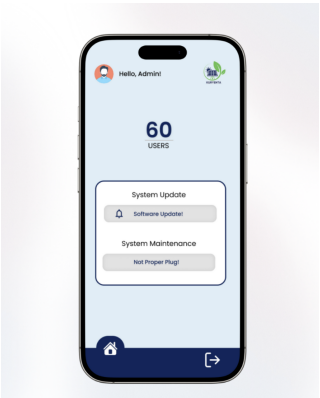
The user dashboard displays a greeting header with location metadata and a card showing the current points balance. A bottom navigation bar provides access to core modules (home, rewards, notifications, and account/tools). This screen serves as the primary hub for user activities.

Figure 3.26
Renting Screen



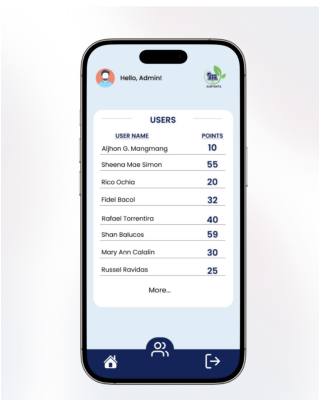
The renting module allows search by Station ID and shows slot availability per station. Users select a slot and view its battery percentage with the corresponding price. Payment options include Pay via Points and Pay via Coin Slot, enabling transaction initiation.

Figure 3.27
Admin Home Screen



The administrative dashboard reports the total number of users and provides status cards for system updates and system maintenance alerts. A bottom navigation enables movement to other administrative functions. This screen supports monitoring and operational control.

Figure 3.28
Admin Manage Users Screen



The user-management view lists registered users with their points bal-

ances. The list supports further exploration through a “More...” action. This screen enables oversight of accounts for incentive tracking and administration.

3.2.11 Materials and Cost

Table 6: Hardware Components and Cost

Component	Price
Battery x 3	P15,600.00
Solar Charge Controller	P1,000.00
Solar Panel	P5,300.00
Pure Sine Wave Inverter x 3	P5,700.00
Lithium Battery Charger x 2	P3,000.00
Low Voltage Disconnect module	P100.00
Coin Slot Facade x 2	P1,000.00
ESP32 x 3	P600.00
DC breaker x 5	P500.00
Flexible Silicone Wire x 2	P270.00
Relay Module 12/ 240v x 2	P240.00
Liquid Crystal Display (LCD) x 2	P400.00
Relay x 3	P600.00
GPS module	P200.00
GSM module	P600.00
AC breaker	P1,200.00
Solenoid lock x 2	P400
Router	P1,500.00
Total	P37,570.00

Table 4 shows the following materials that will be used in the study.

3.2.12 Standards and Guidelines Considered

The design and evaluation of the solar-powered rental station were guided by several internationally recognized standards to ensure safety, re-

liability, and environmental compliance.

- *ISO 9001 (Quality Management)* – Ensures proper documentation, and structured design of the system.
- *ISO 14001 (Environmental Management)* – Supports compliance with sustainable and eco-friendly practices.
- *ISO 9806 (Solar Energy Testing)* – Provides guidelines for evaluating the performance and durability of the solar charging component.
- *ISO 20653 (IP Ratings)* – Used to reference enclosure protection against dust and water for outdoor installation.
- *IEC 62133 (Battery Safety)* – Covers safety requirements for the LifePO4 battery used in the power source.
- *IEC 62368 (Electrical Equipment Safety)* – Provides safety guidelines for electronic components such as the inverter, charge controller, and communication modules.

APPENDICES

Appendix A

Step-by-Step Calculations

Detachable Power Source

The battery capacity for the detachable power source is calculated based on expected daily consumption and system voltage. The battery capacity is calculated as:

$$\begin{aligned}\text{Battery Capacity} &= \frac{\text{Total Daily Consumption}}{\text{System Voltage}} \\ &= \frac{500 \text{ Wh}}{12 \text{ V}} \\ &= 41.67 \text{ Ah} + 0.858 \text{ Ah}\end{aligned}$$

$$\text{Battery Capacity} \approx 43 \text{ Ah}$$

Use: 50Ah LiFePO₄ Battery

Station

The station battery is sized to accommodate the detachable battery and additional reserve. Let BCDPS = Battery Capacity of Detachable Power Source.

The station battery is sized as:

$$\begin{aligned}\text{Battery Capacity} &= (\text{BCDPS} + 35\%) + 0.858\text{Ah} \\ &= 68.358\text{Ah}\end{aligned}$$

$$\text{Battery Capacity} \approx 70\text{Ah}$$

Use: 70Ah LiFePO₄ Battery.

Panel Sizing

Solar panel power is calculated to meet the daily energy requirements within average peak sun hours.

$$\begin{aligned}\text{PV Power} &= \frac{\text{Total Daily Consumption}}{\text{Sun Peak Hours}} \\ &= \frac{500\text{Wh}}{3.5\text{h}} \\ &= 142.857\text{W}\end{aligned}$$

$$\text{PV Power} = 200\text{W}$$

Use: 200W Solar Panel.

Inverter Sizing

The inverter is sized to handle the system load with a safety margin.

$$\text{Inverter Size} = 250\text{W} + 20\% \text{ allowance}$$

$$\text{Inverter Size} = 300\text{W}$$

Use: 300W pure sine wave inverter.

Solar Charge Controller Sizing

The charge controller current is selected based on PV array power and system voltage.

$$\begin{aligned}
 I_{cc} &= \frac{\text{Total PV Power}}{\text{System Voltage}} \\
 &= \frac{200W}{12V} \\
 &= 16.67A
 \end{aligned}$$

$$I_{cc} = 20A$$

Use: 1 pc — MPPT 12V or 24V Auto Adapt 20A.

Breaker Sizing

Circuit breakers are sized to protect each part of the system.

a. Panel - SCC

$$\begin{aligned}
 \text{Circuit Breaker} &= 11.85A \times \text{Over Radiance Factor} \times \text{Safety Factor} \\
 &= 18.5A
 \end{aligned}$$

$$\text{Circuit Breaker} = 20A$$

b. SCC - Battery

$$\begin{aligned}\text{Circuit Breaker} &= \frac{\text{Total Array Wattage}}{\text{Battery Bank Nominal Voltage}} \times \text{Safety Factor} \\ &= \frac{200W}{12V} \times 1.25\end{aligned}$$

$$\text{Circuit Breaker} = 20.8A$$

c. Battery - Power Inverter

$$\begin{aligned}\text{Circuit Breaker} &= \frac{\text{Power Inverter}}{\text{Power Bank Nominal Voltage}} \\ &= \frac{1000W}{12V} \\ &= 83.33A\end{aligned}$$

$$\text{Circuit Breaker} = 80A$$

d. Inverter - Load (AC Breaker)

BBNV = Battery Bank Nominal Voltage

$$\begin{aligned}\text{Circuit Breaker} &= \frac{\text{Safety Load}}{\text{Inverter Voltage Output}} \times \text{BBNV} \\ &= \frac{700W}{220V} \times 1.25V\end{aligned}$$

$$\text{Circuit Breaker} = 3.98A$$

Wire Sizing

Wire sizes are determined to handle current and limit voltage drop.

a. Panel - SCC

$$\begin{aligned}\text{VDI} &= \frac{10.85 \times 12}{18.5 \times 2} \\ &= 3.53\end{aligned}$$

Wire Size = 10*AWG*

b. SCC - Battery

$$\begin{aligned}\text{VDI} &= \frac{20 \times 4}{12.8 \times 2} \\ &= 3.125\end{aligned}$$

Wire Size = 10*AWG*

References